

السلام عليكم ورحمة الله وبركاته

INSTRUCTIONS FOR STUDENTS

- Download The Course Material And The Assignments From Moodle Site
- Use These Materials To Answer Assignments, Mid Term, And Final Exam
- Prepare The Assignments Answer As Pdf, Write Your Name And Section At The End
- Send Assignments And Tasks To Dr **What up**

Evolution of Data Computing SystemsDCS

Distributed Data Processing DCS Definition:

- As the volume and complexity of data continue to grow, traditional centralized and decentralized data processing methods face serious challenges.
- Organizations and Companies trying to extract valuable insights from huge data sets need efficient and scalable processing capabilities to be able to make efficient decisions at scale.
- One of these capabilities is DCS.
- This Course discusses the fundamental concepts of DCS, including its advantages, how it works, and its role in contemporary data analysis and decision-making.
- DCS refers to the approach of handling and analyzing data across multiple interconnected devices or nodes.
- In contrast to centralized data processing, where all data operations occur on a single, powerful system, DCS decentralizes these tasks across a network of computers.
- This method leverages the collective computing power of interconnected devices, **enabling parallel and clustering processing, middleware, virtualization and faster data analysis**.

Benefits of DCS

1. Scalability:

One of the primary advantages of DCS is scalability.

As data volumes grow, organizations can expand their processing capabilities by adding more nodes to the network.

This scalability ensures that the system can handle increasing workloads without a significant drop in performance, providing a flexible and adaptive solution to the challenges

2. Fault Tolerance:

DCSs inherently offer improved fault tolerance compared to centralized systems.

In a distributed environment, if one node fails, the remaining nodes can continue processing data, reducing the risk of a complete system failure.

This resilience is crucial for maintaining uninterrupted data operations in mission-critical applications.

3. Performance:

Parallel processing, a key feature of DCS, contributes to enhanced performance.

By breaking down complex tasks into smaller subtasks distributed across nets, the system can process data more quickly and efficiently.

This results in reduced processing times and improved overall performance, enabling organizations to derive insights from data in a timely manner.

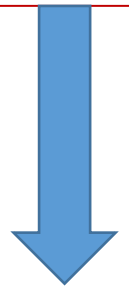
Benefits of DCS

4. Efficient Handling of Large Volumes of Data:

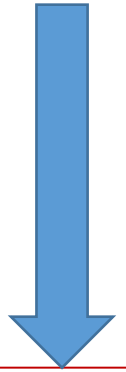
- In the era of big data, efficiently handling large volumes of data is a paramount concern for organizations.
- DCS cope with this aspect by employing data partitioning strategies.
- Large data sets are divided into smaller, more manageable segments, and each segment is processed independently across distributed nodes.
- This approach not only accelerates data processing through parallelism and clustering but also optimizes use of resources. Each node focuses on a specific subset of the data, ensuring that the system operates efficiently and effectively.
- The ability to efficiently handle large volumes of data positions organizations to extract meaningful insights, identify patterns, and make informed decisions.

Evolution of Data Computing Systems DCS

1. Computer Centers



2. Management Information System MIS



3. Data Centers

Evolution of DCSs

1. Computer Centers





Roles in Computer Centers



Application Developer



Production Control Analyst



End User

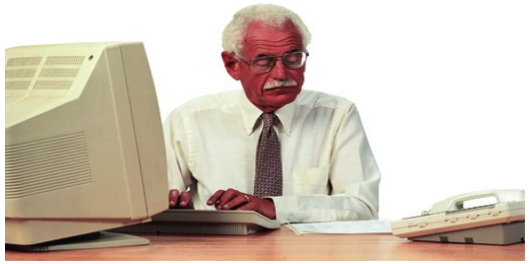




Operator

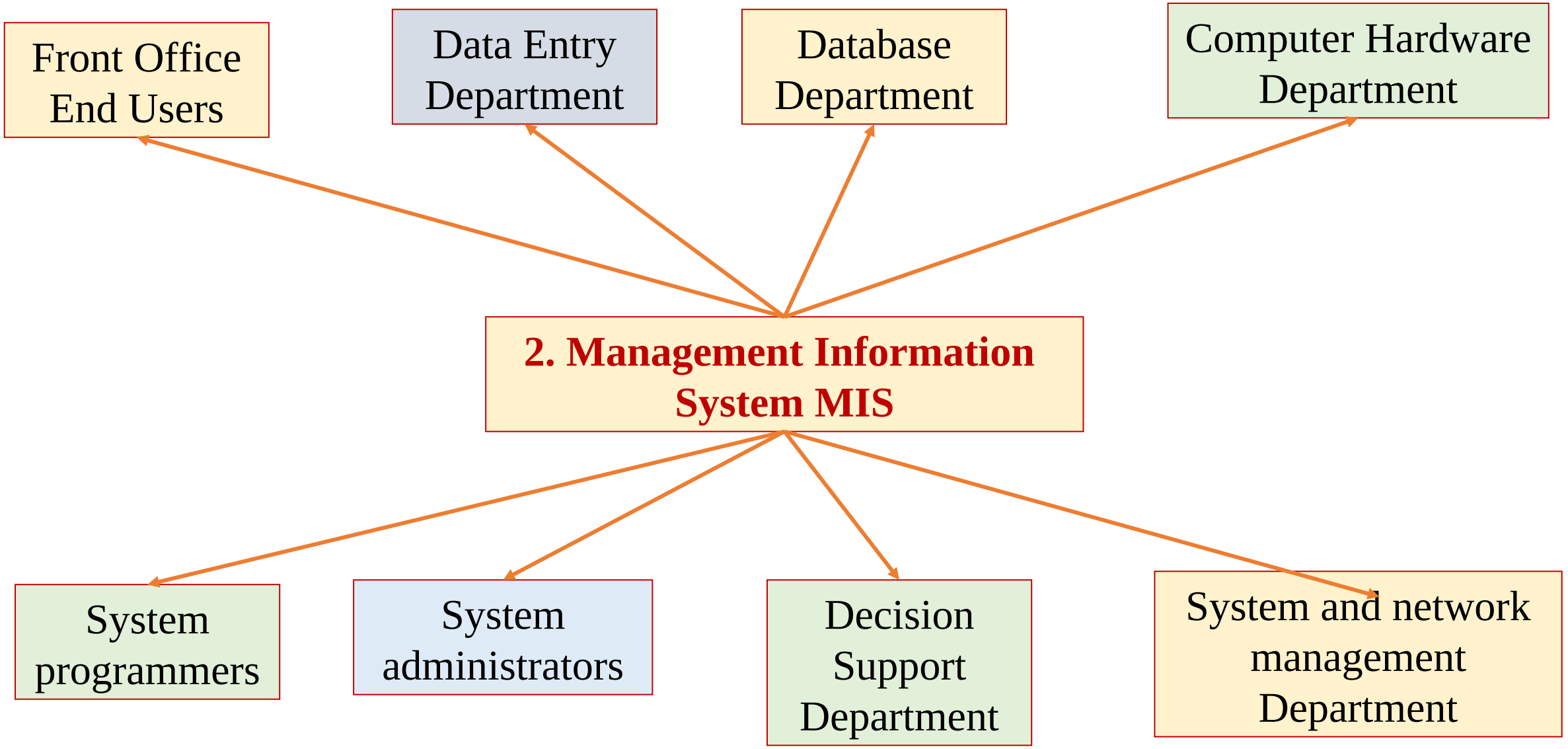


System Programmer



System Administrator

Evolution of Data Computing SystemsDCS



The MIS Jobs

IT jobs fall into the following categories:

- Operations and help desk
- Programming
- Systems design
- Cybersecurity Engineers
- System Analysis
- Web design and Web hosting
- Network design and maintenance
- Database design and maintenance
- Robotics and artificial intelligence



CTO/CIO

- The top job in the information systems field belongs to either the C**hief Technology Officer (CTO) or the Chief Information Officer (CIO)**
- This person is responsible for long-range information systems planning and for keeping abreast of new developments in the field that can affect a company's productivity or competitiveness

Manager of Information Systems Services

This person is responsible for managing all the hardware, software, and personnel within the information systems department

Systems Analyst

Responsible for the design and implementation of information systems
Knowledge of information systems background, this position requires a thorough understanding of business systems and functional areas within a business organization

Cyber Security system Engineer

Implement and monitor security measures for the protection of computer systems, networks, and information. Identify and define defensive and offensive system security requirements.
Design computer security architecture and develop detailed cyber security designs.

Network Administrator

- Administrates internal and external network systems, designing and implementing network systems that deliver correct information to the right decision-maker in a timely manner
- Providing network and cybersecurity is part of this position's responsibility

Database Administrator

- DBA is responsible for database design and implementation
- DBA should have knowledge and understanding of data warehouses and data mining tools

Computer Programmer

- A computer programmer writes computer programs or software pieces that allow the information system to perform a specific task
- There are many computer languages available, and each one requires a specific knowledge suitable for a specific application

Webmaster

- A webmaster designs and maintains the organization's Web site
- Because of the popularity of e-commerce applications, webmasters have been in high demand

Data Centers And Cloud Computing Technologies

3. Data Centers

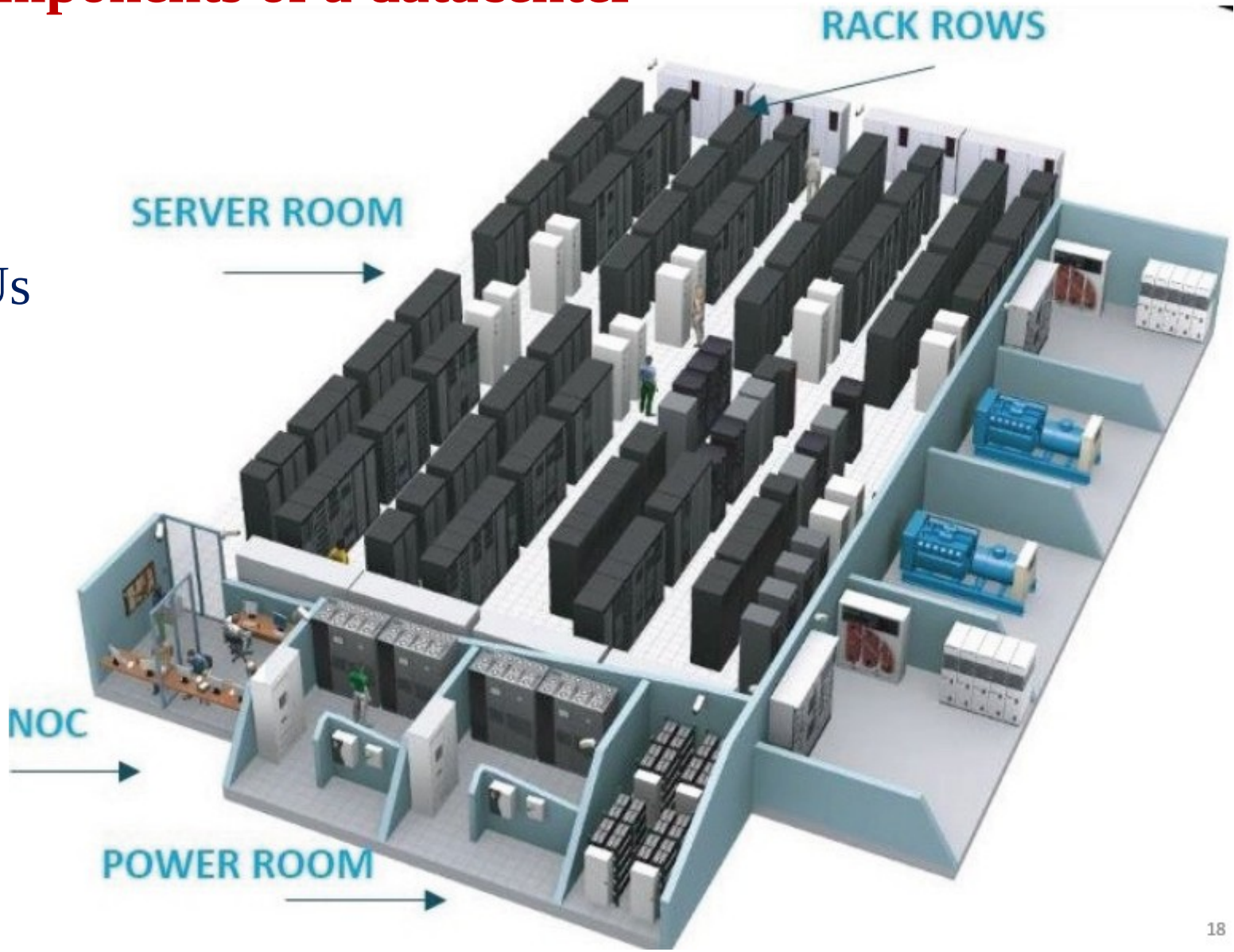


Datacenter Definition

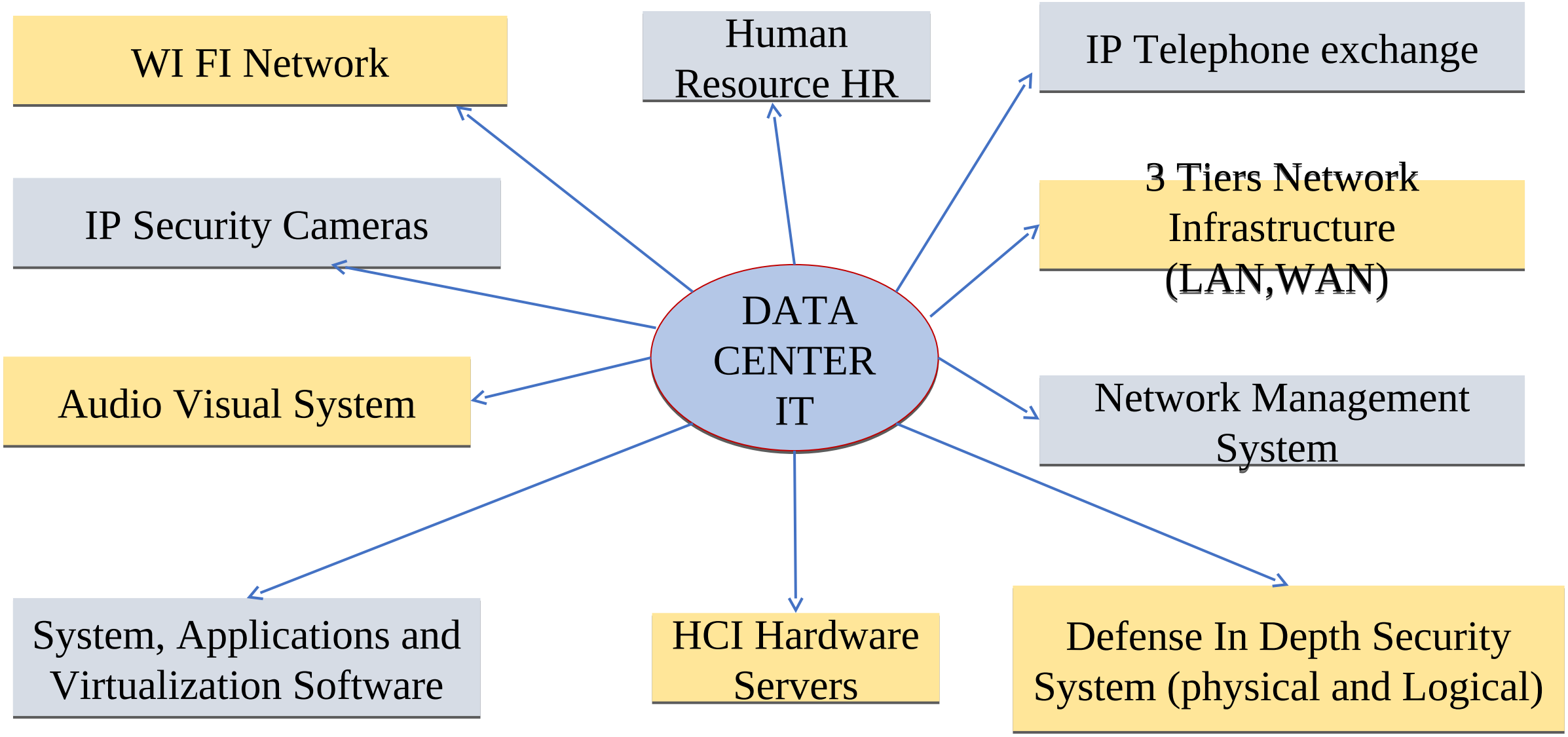
- In today's digital age, data has become an essential asset for businesses, governments, organizations, and individuals.
- Data Centers DCs are the physical facilities that house the IT equipment and infrastructure necessary to store, process, and distribute vast and ever-growing volume of data.
- The housed and supported IT equipment including: servers, storage systems, network devices, cybersecurity devices, and other infrastructure components.
- DCs provide the physical infrastructure and environmental conditions necessary for the secured, reliable and efficient operation of these IT systems.
- DCs are typically designed with multiple levels of redundancy to ensure that they can continue to operate even in the event of a power outage, hardware failure, or other disruption.
- They employ advanced security physical and logical measures to protect the data and equipment from unauthorized access and cyber threats.

Main components of a datacenter

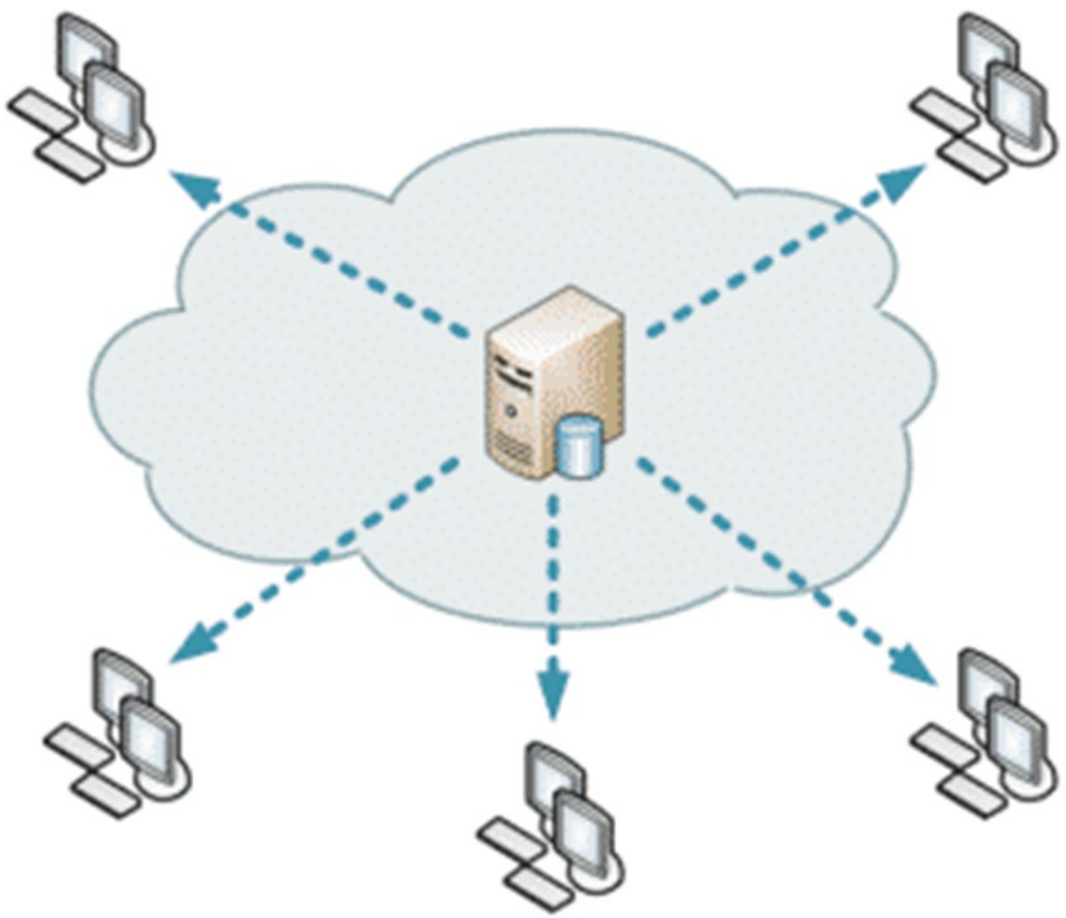
1. Human Resources
2. Physical Suitable Site
3. Racks and enclosures
4. Power Distribution Units PDUs
5. Cooling systems
6. Fire suppression systems
7. Security systems
8. Computer resources
9. Storage systems
10. Network equipment
11. Software and virtualization
12. Data Center Management



DATA CENTER IT COMPONENTS



CMP521/CMPN421/CMPn421: Distributed Computer Systems DCS



Centralized Computer Systems



Distributed Computer Systems

Course Contents

This course provides an overview of the principles and concepts of DCSs technologies (middleware, clustering computing, virtualization, cloud computing and data center), applications, details of the hardware, software architecture, security, and communications support required.

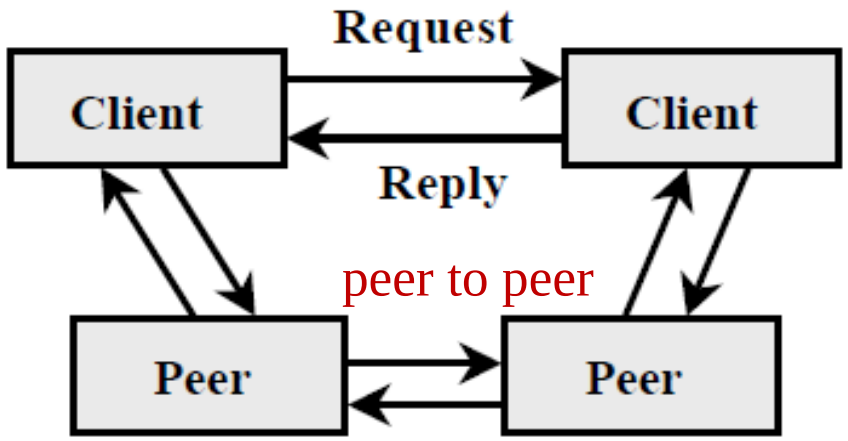


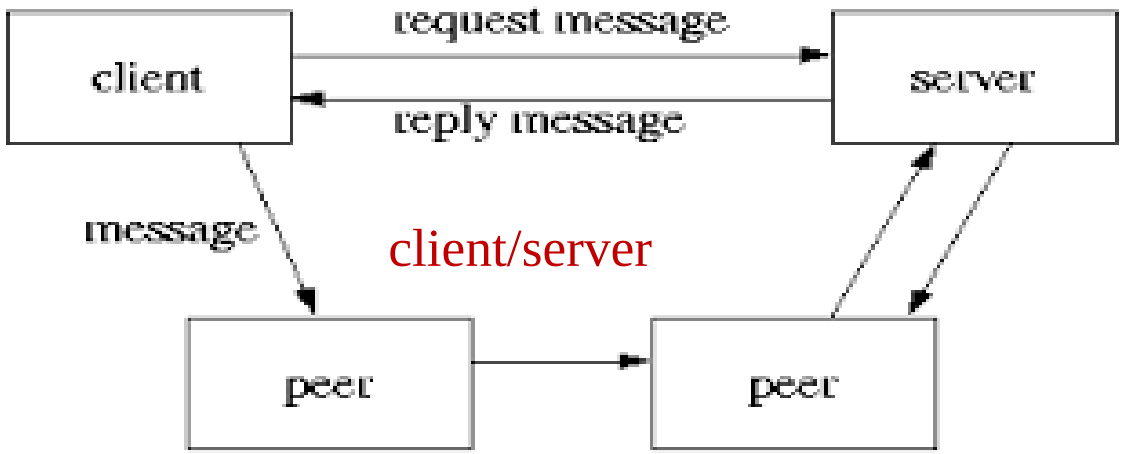
Course Contents

1. Introduction to Distributed Computer Systems DCS
2. Introduction to Distributed Information Systems DIS
3. DCS models and examples
4. DCSs Types
5. DCSs architectures
6. DCSs storage systems (SAS, NAS, and SAN)
7. DCSs high performance computing HPC
8. Data Centers and Cloud Computing
9. **Distributed Data Processing Systems DDP**
10. **Distributed database Systems**
11. Distributed Computer Systems Security

1.1. Course Aims:

- All large ISs are distributed. E.g. enterprise wide business systems must support multiple users running common applications across different distributed computing sites.
- A DCS contains a variety of applications that utilize DCS underlying support software, the hardware they run on, and the network connecting them.
- The largest and best-known DCS is the set of computers, software, and services comprising the WWW (Facebook, google, YouTube,…), which is so pervasive that it coexists with and connects to most other existing DCSs.
- The common DCS are networked peer to peer and client/server systems.





DCS Course Syllabus

Purpose: To convey insight into principles underlying the applications and design of DCSs and providing students with the necessary tools for evaluating DCS, designing their basic components, and how to secure them.

Expected Learning Outcomes

At the end of the semester, the students will be able:

- Understand DCSs environments (characteristics and advantages)
- Study state of the art and practice in DCS
- Assess problems related to DCS environments and how to solve them.

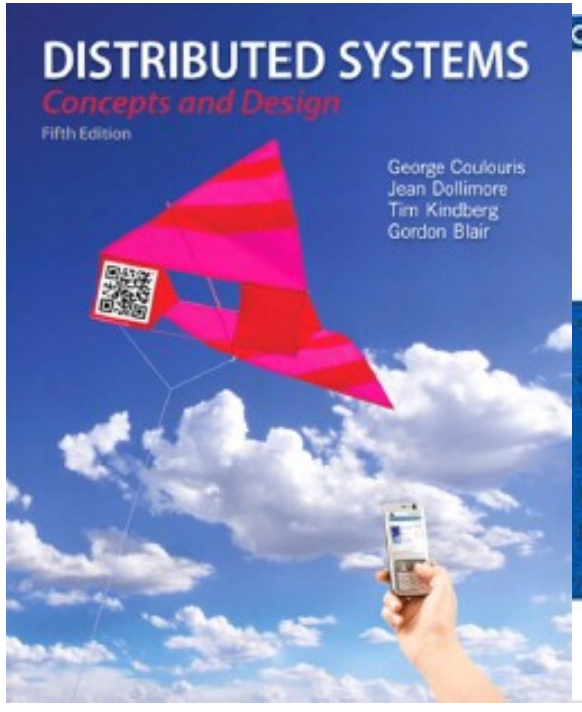
The course has three main goals:

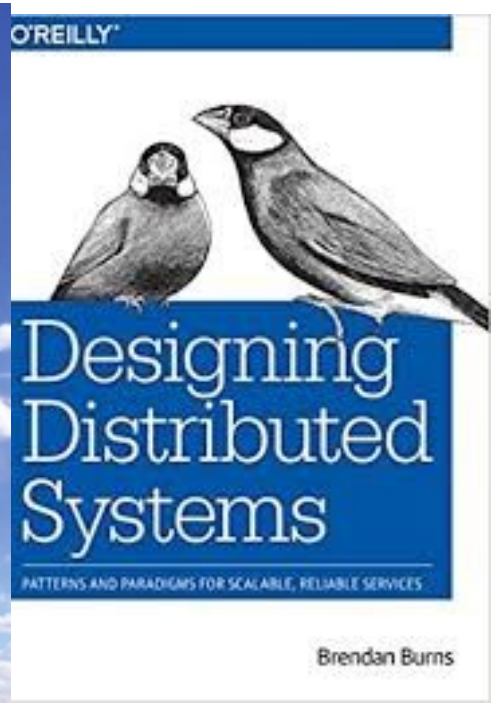
- To learn the principles, and architectures models used in DCSs.
- To examine state-of-the-art examples of DCSs
- To design and evaluate sample of DCS.

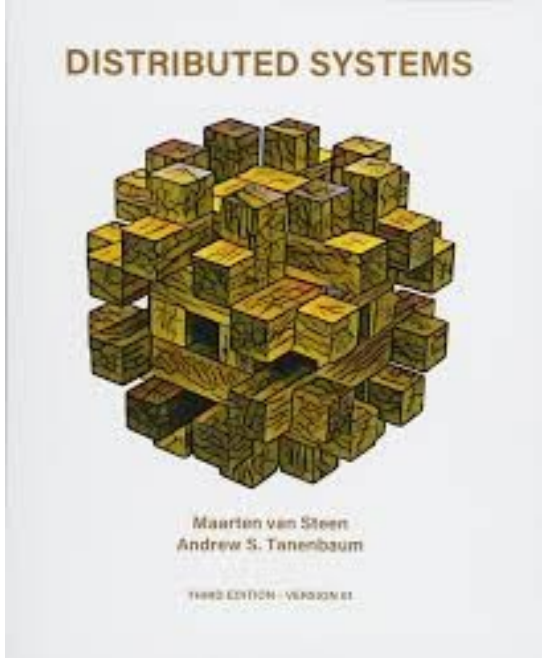
Through these objectives, the course will transform Students to computational thinking from designing applications for a single computer system, towards that of DCS.

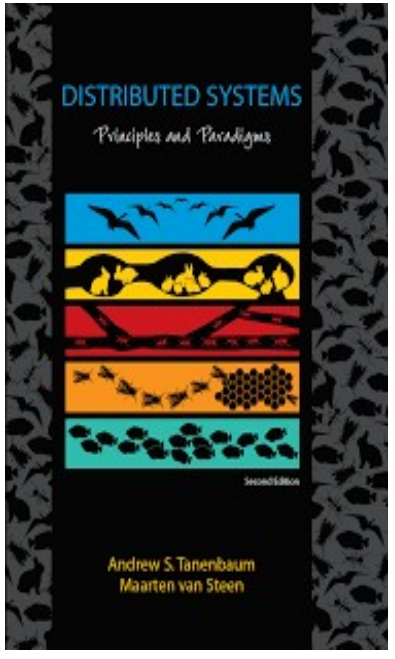
Textbooks: The primary textbooks for this course are:

- Andrew S. Tannenbaum and Maarten Van Steen, Distributed Systems: Principles and Paradigms, Second Edition, Pearson, 2007.
- George Coulouris, Jean Dollimore, Tim Kinberg, and Gordon Blair Distributed Systems: Concepts and Design, Fifth Edition, Addison Wesley, 2011.
- James E. Smith, and Ravi Nair Virtual machines: Versatile Platforms for Systems and Processes, First Edition, Morgan Kaufmann, 2005.







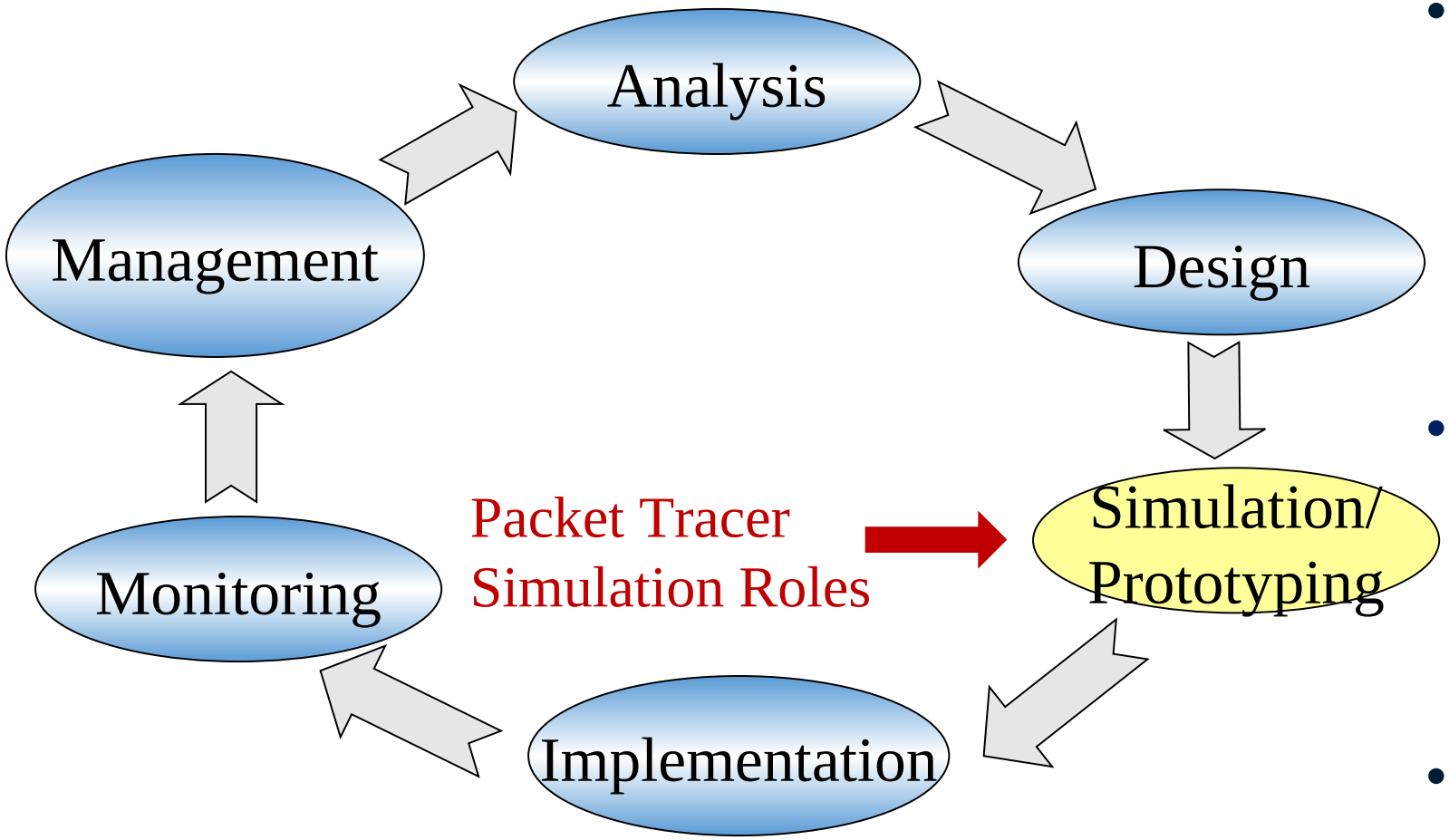




Grading/Evaluation Law 2018/Law 2020: LAB

Packet Tracer 8.2.2. Activity	م
Packet Tracer 8.2 Simulation: Downloading, Installation and features	1
Peer to Peer and Server Based Networks	2
HTTP AD HTTPS SERVERS	3
AAA SERVER	4
DNS SERVER	5
FTP AND TFTP SERVER	6
DHCP SERVER	7
MAIL SERVER	8
Implementing Secured Wired and Wireless Nets	9

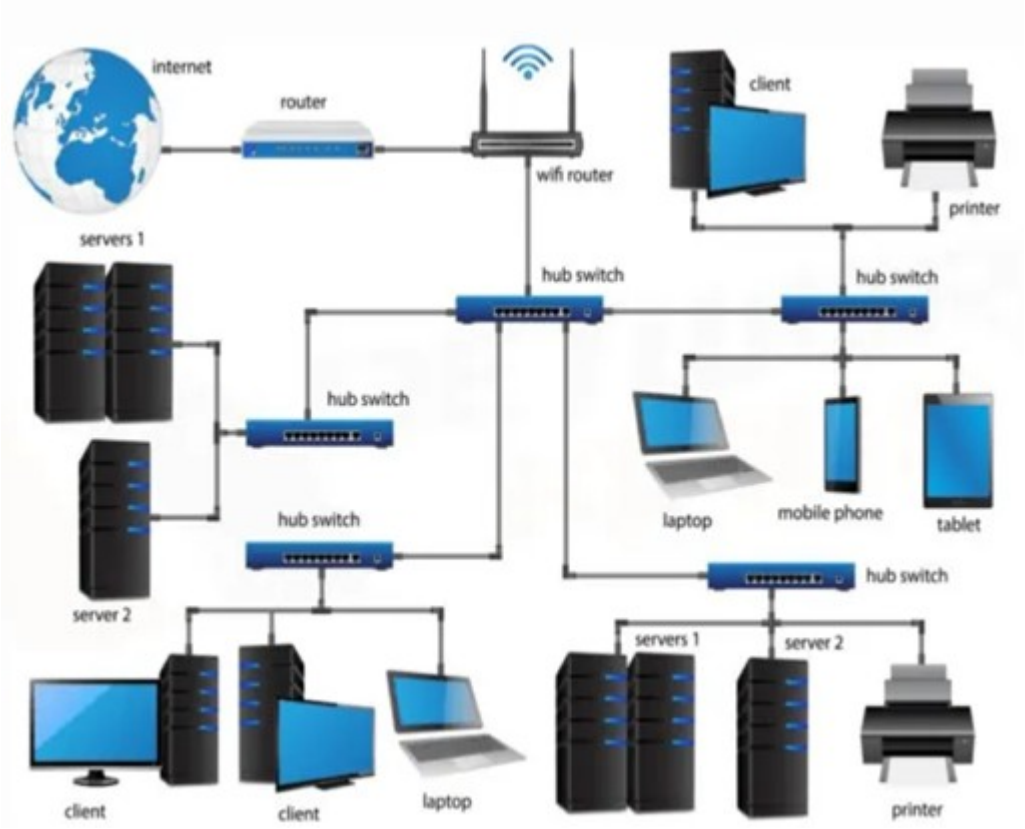
Network Design and Development Life Cycle



- Cisco Packet Tracer is a network simulation tool that allows users to create virtual network topologies, configure devices, and simulate network behavior.
- It's used for learning and practicing networking concepts, troubleshooting network configurations, and visualizing network protocols.
- It offers a user-friendly interface and a wide range of features for both beginners and experienced users.

Features of the Cisco Packet Tracer

- Simulates network devices and protocols
- Supports multiple protocols and technologies
- Provides a comprehensive set of networking tools
- Includes a range of network and security devices and topologies
- Allows for multi-user collaboration



DCS whose hardware applications, networking, and security are designed, simulated, and tested using the Packet Tracer

Course Grading Information LAW 2018/2020

Code	Subject	Total Credits	L	Contact Hours		prerequisite
				T	P	
CMPN421/	**Distributed Computer**	**3**	**2**	**2**	**-**	**CMPN421/**

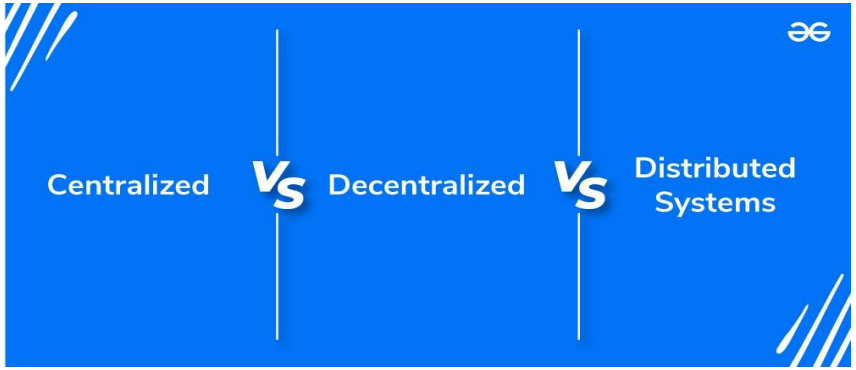
Assessment Method	Timing	Grade (Degrees)
Year Work: seminars, quizzes assignments and reports	Bi-Weekly	20
Mid-Term Exam	6-th Week	20
Practical Exam	LAB	20
Written Exam	16-th week	40
Total		100

Course Grading Information LAW 2012

Code	Subject	Total Credits	L	Contact Hours		prerequisite
				T	P	
CMP 521	Distributed Computer Systems	3	2	2	-	CMP 421

Assessment Method	Timing	Grade (Degrees)
Year Work: seminars, quizzes assignments and reports	Bi-Weekly	20
Mid-Term Exam	6-th Week	10
Practical Exam	Fifteenth week	-
Written Exam	Sixteenth week	70
Total		100

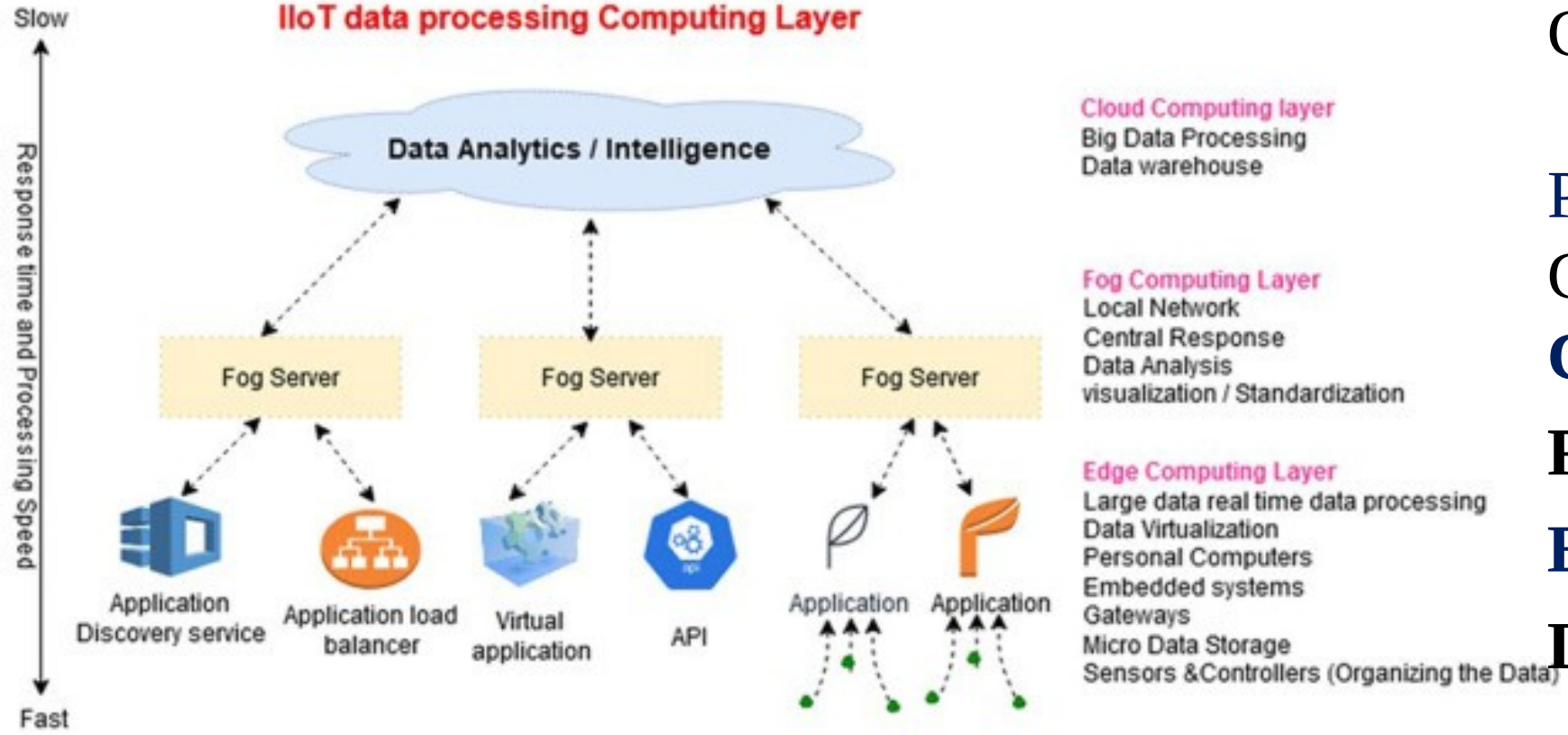
Centralized vs. Decentralized vs. Distributed Systems



- Understanding the systems architecture is necessary for designing efficient and effective solutions.
- Centralized, decentralized, and distributed systems each offer unique advantages and challenges.
- Centralized systems rely on a single point of control, providing simplicity but risking a single point of failure.
- Decentralized systems distribute control among multiple nodes, enhancing fault tolerance and scalability.
- Distributed systems further spread resources across multiple locations, optimizing performance and reliability.

Evolution of Computing Systems

Centralized Computing System CC → Decentralized Computing Systems DC → **Distributed Computing Systems DCS**



Client-server architecture.

Peer-to-peer architecture.

Grid computing.

Cloud computing.

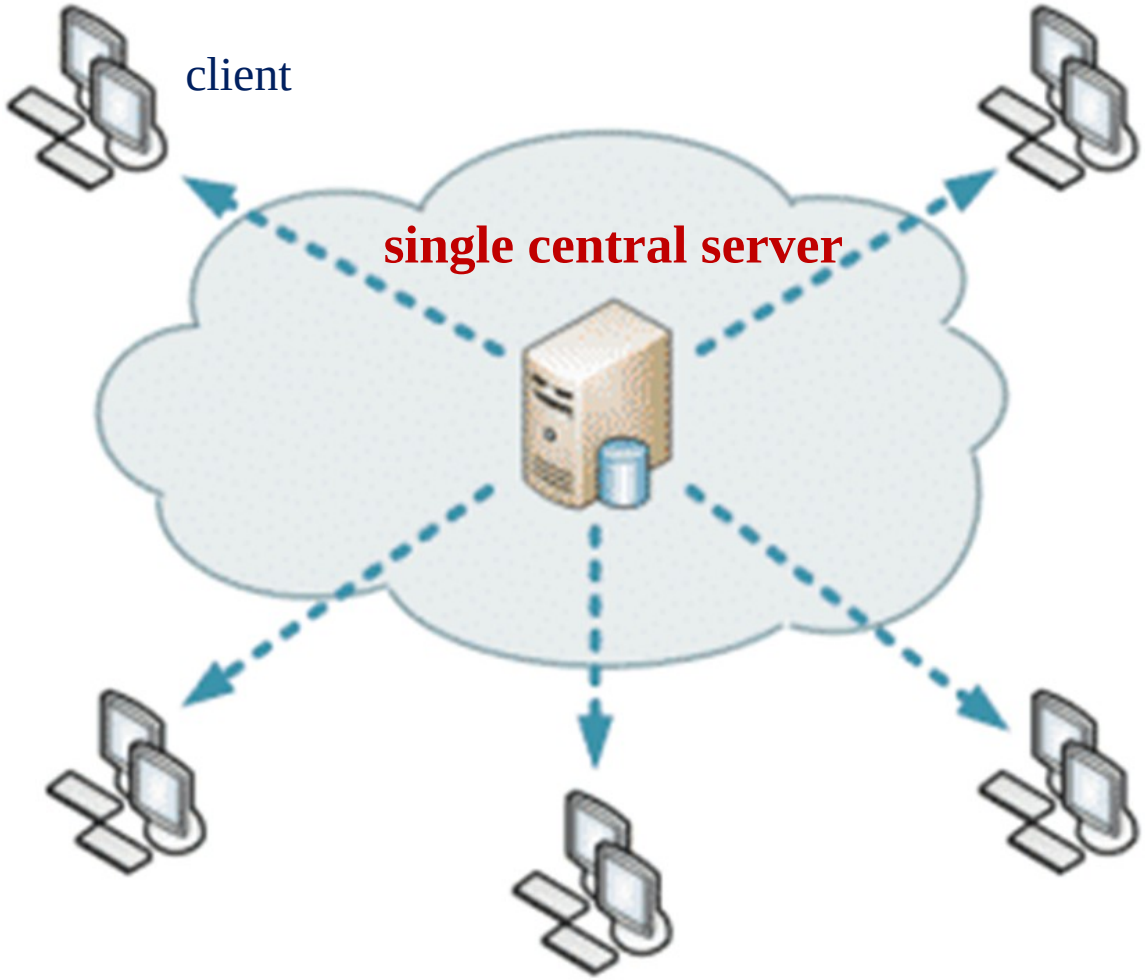
Fog computing

Edge computing.

Distributed databases

Centralized Systems

- Centralized Systems are a type of computing architecture where all or most of the processing and data storage is done on a single central server.
- This central server manages all operations, resources, and data, acting as the hub through which all client requests are processed.
- The clients, or nodes, connected to the central server typically have minimal processing power and rely on the server for most computational tasks.



Key Characteristics of Centralized Systems

ADV.

- **Single Point of Control**: All data processing and management tasks are handled by the central server. Easier to manage and maintain since there is one primary location for administration.
- **Simplicity**: Simplified architecture with a clear structure where all operations are routed through the central node. Easy to deploy and manage due to centralized nature.
- **Efficiency**: Efficient use of resources as the central server can be optimized for performance. Easier to implement security measures and updates centrally.

DISADV.

Scalability Issues: Limited scalability as the central server can become a bottleneck if the load increases. Adding more clients can strain the server's resources, leading to performance degradation.

Single Point of Failure: If the central server fails, the entire system can become inoperative. High availability and redundancy measures are essential to mitigate this risk.

Centralized Systems use cases

Enterprise Resource Planning (ERP) Systems:

- **Description**: Centralized ERP systems manage and integrate core business processes such as finance, HR, and supply chain in a single system.
- **Benefits**: Simplified management, consistent data, and centralized control over business processes.

Customer Relationship Management (CRM) Systems:

- **Description**: Centralized CRM systems store and manage customer data, interactions, and sales processes in one location.
- **Benefits**: Improved customer data consistency, streamlined customer service, and centralized reporting.

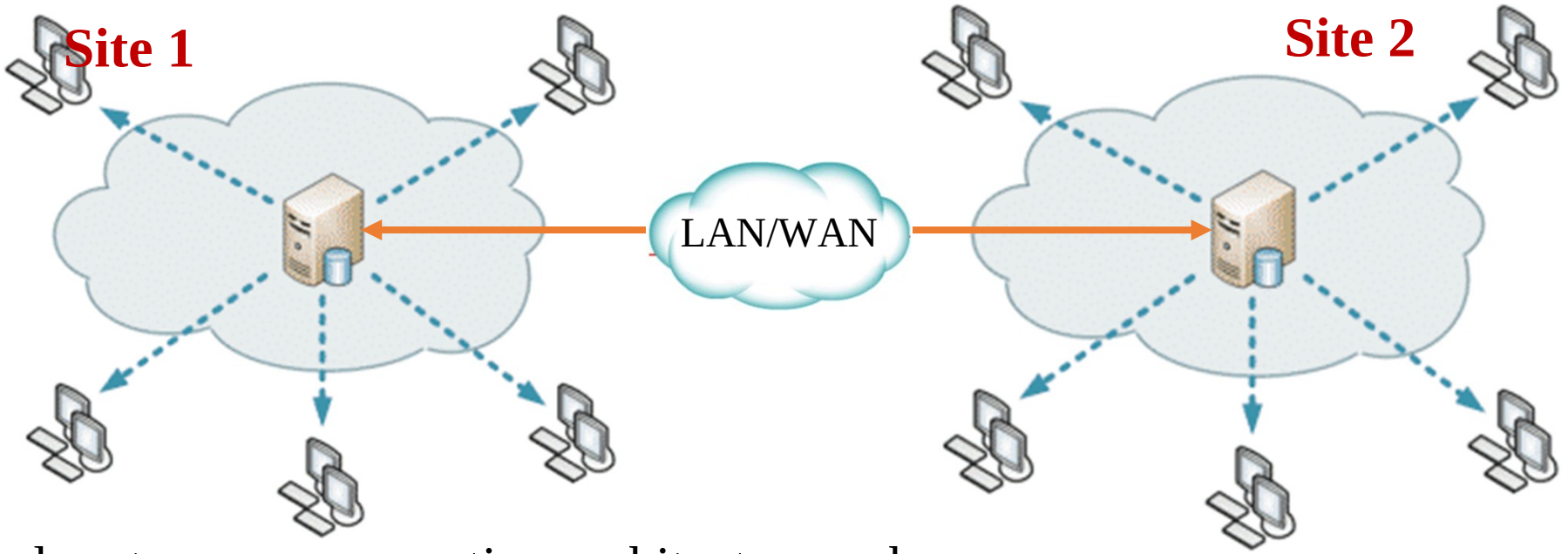
Email Servers:

- **Description**: Centralized email servers manage and store email communications for an organization.
- **Benefits**: Centralized email storage, simplified Desktop and security measures, and easy management of user accounts.

Banking Systems:

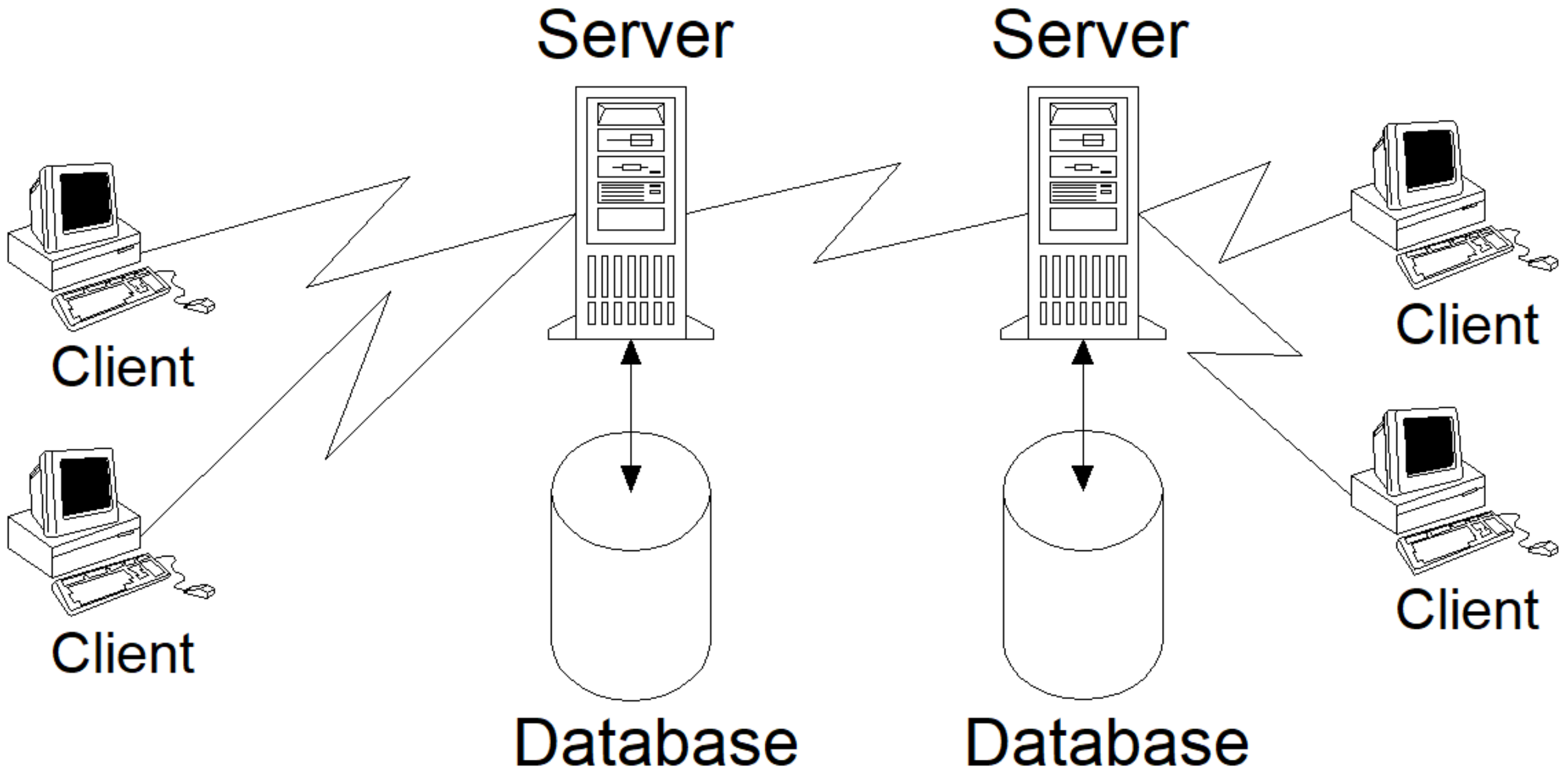
- **Description**: Centralized banking systems manage customer accounts, transactions, and financial services through a central server.
- **Benefits**: Enhanced security, centralized transaction processing, and consistent financial records.

Decentralized Systems



- Decentralized systems are computing architectures where multiple nodes, spread across different locations/ sites, share control and processing power without a single central authority.
- Each node in a decentralized system operates independently but collaborates with others to achieve common goals.
- This structure enhances fault tolerance, scalability, and resilience compared to centralized systems.

Decentralized Systems



Key Characteristics of Decentralized Systems:

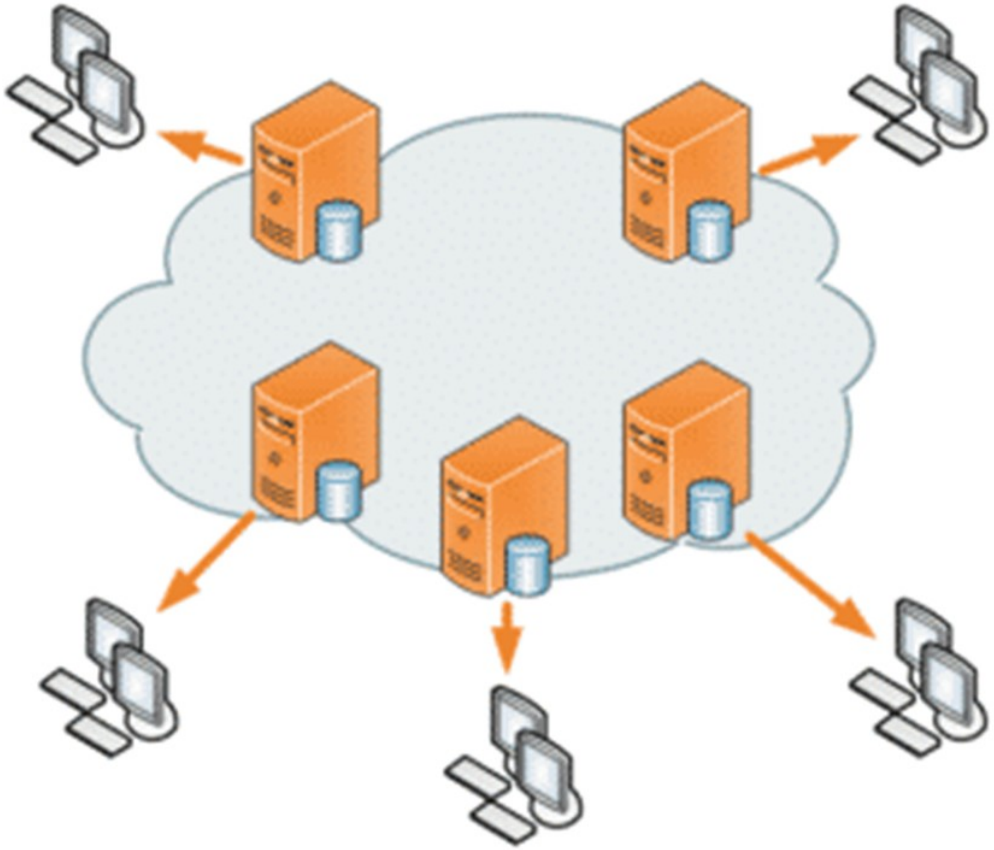
- **Distributed Control**:
  - No single point of control or failure.
  - Each node operates independently, contributing to the overall system's functionality.
- **Fault Tolerance**:
  - If one node fails, the system can continue to function with the remaining nodes.
  - Enhanced resilience against failures and attacks.
- **Scalability**:
  - Easier to scale by adding more nodes without overwhelming a central point.
  - Load distribution across multiple nodes improves performance and resource utilization.
- **Coordination and Communication**:
  - Nodes must communicate and coordinate to maintain system integrity and consistency.
  - Complex algorithms and protocols often manage this coordination.
- **Autonomy and Redundancy**:
  - Each node can operate autonomously, contributing to redundancy and reducing single points of failure. Data and services are replicated across multiple nodes for reliability.

Decentralized system use cases

- **Peer-to-Peer (P2P) File Sharing**: Networks where users share files directly with each other without a central server.
  - **Benefits**: Increased resilience, reduced central bottlenecks, and distributed resource sharing.
  - **Examples**: BitTorrent, Gnutella.
- **Decentralized Finance (DeFi) Platforms**: Financial services built on blockchain technology, offering services like lending, borrowing, and trading without intermediaries.
  - **Benefits**: Greater accessibility, reduced fees, and increased transparency.
- **Mesh Networks**: Networks where each node relays data for the network, providing a decentralized approach to internet connectivity.
  - **Benefits**: Increased network resilience, scalability, and coverage in remote areas.
  - **Examples**: Community-based Wi-Fi networks, disaster recovery networks.

Distributed systems

- Distributed systems are computing architectures where multiple independent nodes or computers work together to achieve a common goal.
- These nodes communicate and coordinate with each other over a network, appearing as a single coherent system to the end user.
- Distributed systems aim to improve performance, reliability, scalability, and resource sharing by leveraging the collective power of interconnected devices.



Key Characteristics of Distributed Systems

- **Geographical Distribution**:
  - Nodes are spread across different physical locations.
  - They communicate via a network, such as a local area network (LAN) or the internet.
- **Resource Sharing**:
  - Nodes share resources such as processing power, storage, and data.
  - This enables more efficient utilization of resources.
- **Concurrency**:
  - Multiple nodes operate concurrently, performing tasks simultaneously.
  - This parallelism enhances the system's overall performance and throughput.
- **Scalability**:
  - Easy to scale by adding more nodes to the system.
  - System capacity and performance improve with the addition of resources.
- **Fault Tolerance** :
  - Designed to handle failures gracefully.
  - Redundancy and replication ensure the system remains operational even if some nodes fail.
- **Transparency**:
  - The complexity of the distributed system is hidden from users.
  - Users interact with the system as if it were a single entity.

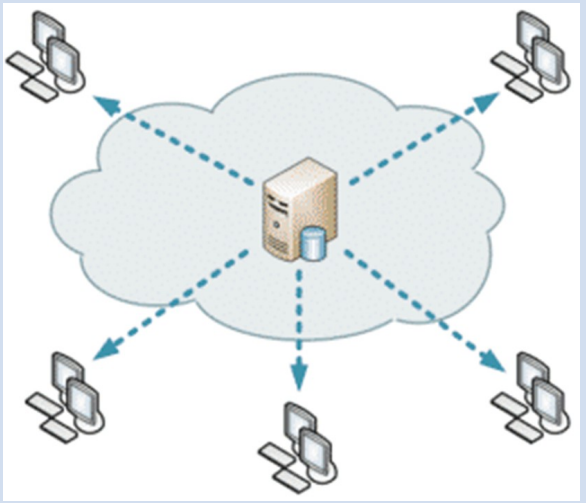
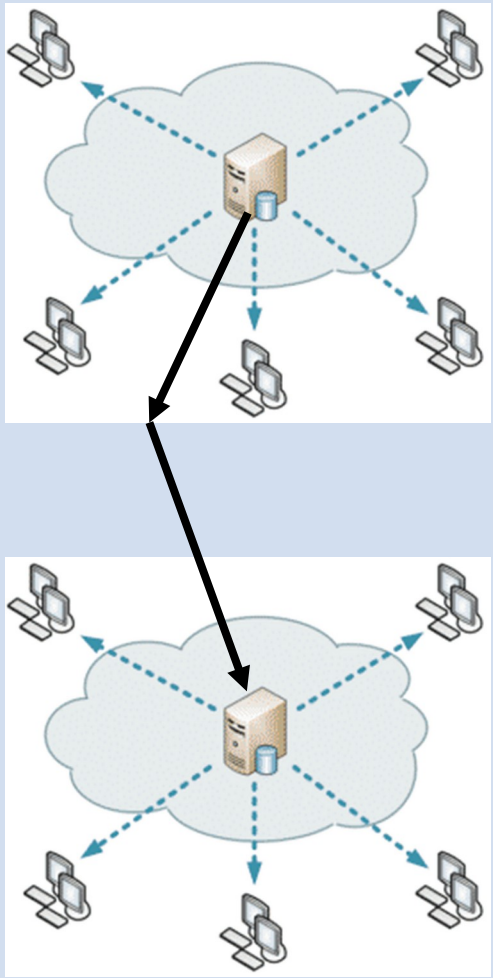
DCS use cases

- **Cloud Computing Platforms**:
  - **Description**: Cloud services provide scalable and flexible computing resources over the internet.
  - **Benefits**: On-demand resource allocation, high availability, and fault tolerance.
  - **Examples**: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform.
- **Content Delivery Networks (CDNs)**:
  - **Description**: Networks of distributed servers that deliver web content based on users' geographic locations.
  - **Benefits**: Reduced latency, increased content delivery speed, and load balancing.
  - **Examples**: Akamai, Cloudflare, Amazon CloudFront.
- **Distributed Databases**:
  - **Description**: Databases that store data across multiple servers to improve performance and reliability.
  - **Benefits**: High availability, scalability, and fault tolerance.
  - **Examples**: Google Spanner, Amazon DynamoDB
- **Microservices Architectures**:
  - **Description**: Architectures where applications are built as a collection of loosely coupled services.
  - **Benefits**: Improved scalability, easier maintenance, and fault isolation.
  - **Examples**: Netflix, Uber, Amazon.

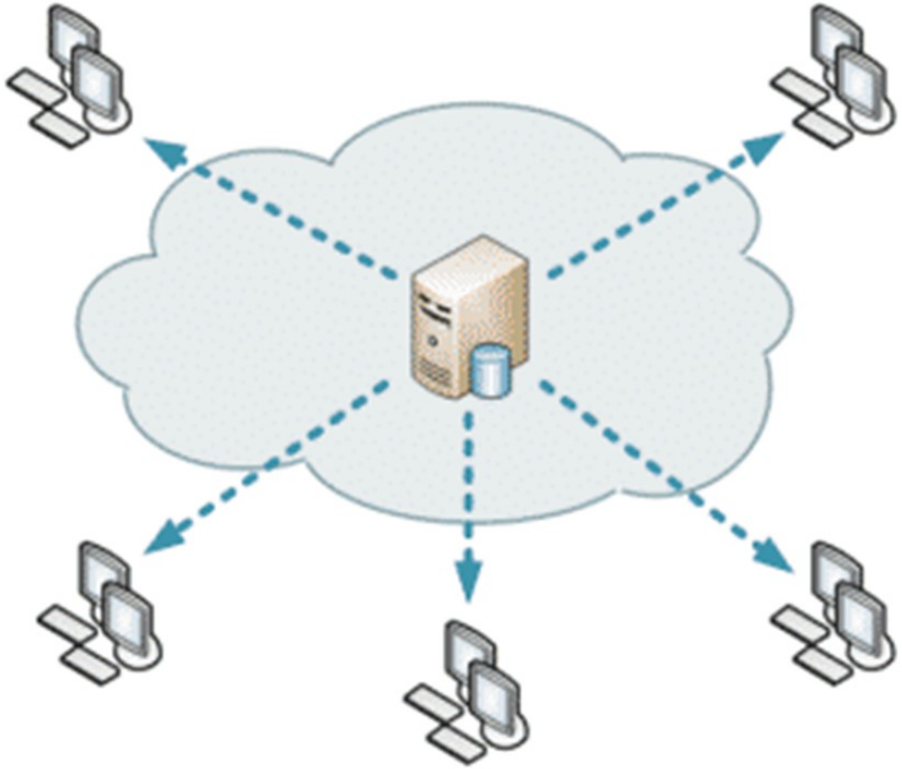
Differences between Centralized, Decentralized and Distributed Systems

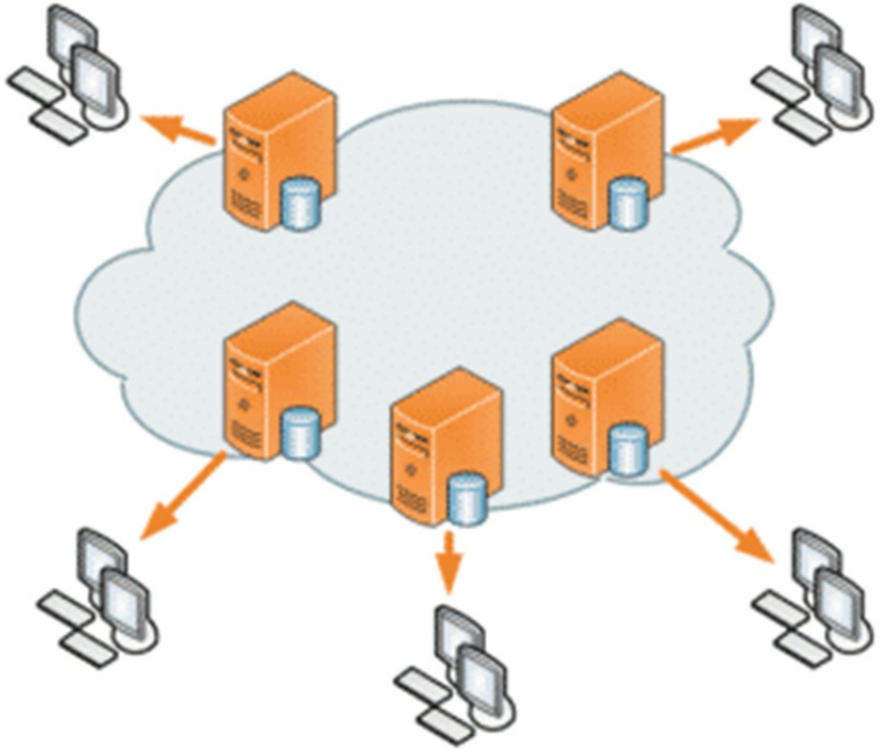
Aspect	Centralized Systems	Decentralized Systems	Distributed Systems
Definition	Single central server controls and manages all operations.	Multiple nodes with independent control, no central authority.	Multiple interconnected nodes working together as a single system.
Control	Centralized control with a single point of management.	Distributed control, each node operates independently.	Shared control, nodes collaborate to achieve common goals.
Single Point of Failure	High risk; if the central server fails, the whole system fails.	Reduced risk; failure of one node does not impact the entire system.	Reduced risk; designed for fault tolerance and redundancy.
Scalability	Limited scalability, can become a bottleneck.	More scalable, can add nodes independently.	Highly scalable, can add more nodes to distribute the load.
Resource Utilization	Central server resources are heavily utilized.	Resources are spread across multiple nodes.	Efficient resource sharing across nodes.
Performance	Can be high initially but may degrade with increased load.	Generally good, performance improves with more nodes.	High performance due to parallel processing and resource sharing.
Management	Easier to manage centrally.	More complex, requires managing multiple nodes.	Complex, requires coordination and management of many nodes.
Latency	Lower latency, as operations are managed centrally.	Can vary, depends on the distance between nodes.	Potentially higher latency due to network communication.

Differences between Centralized, Decentralized and Distributed Systems

Aspect	Centralized Systems	Decentralized Systems	Distributed Systems
Drawing			

Centralized VS DCS Summary





Centralized Computer System Characteristics

- One component with non-autonomous parts
- Component shared by users all the time
- All resources accessible
- Software runs in a single process
- Single Point of control (technical support+ development)
- Single Point of failure/solved by redundancy

Distributed Computer System Characteristics

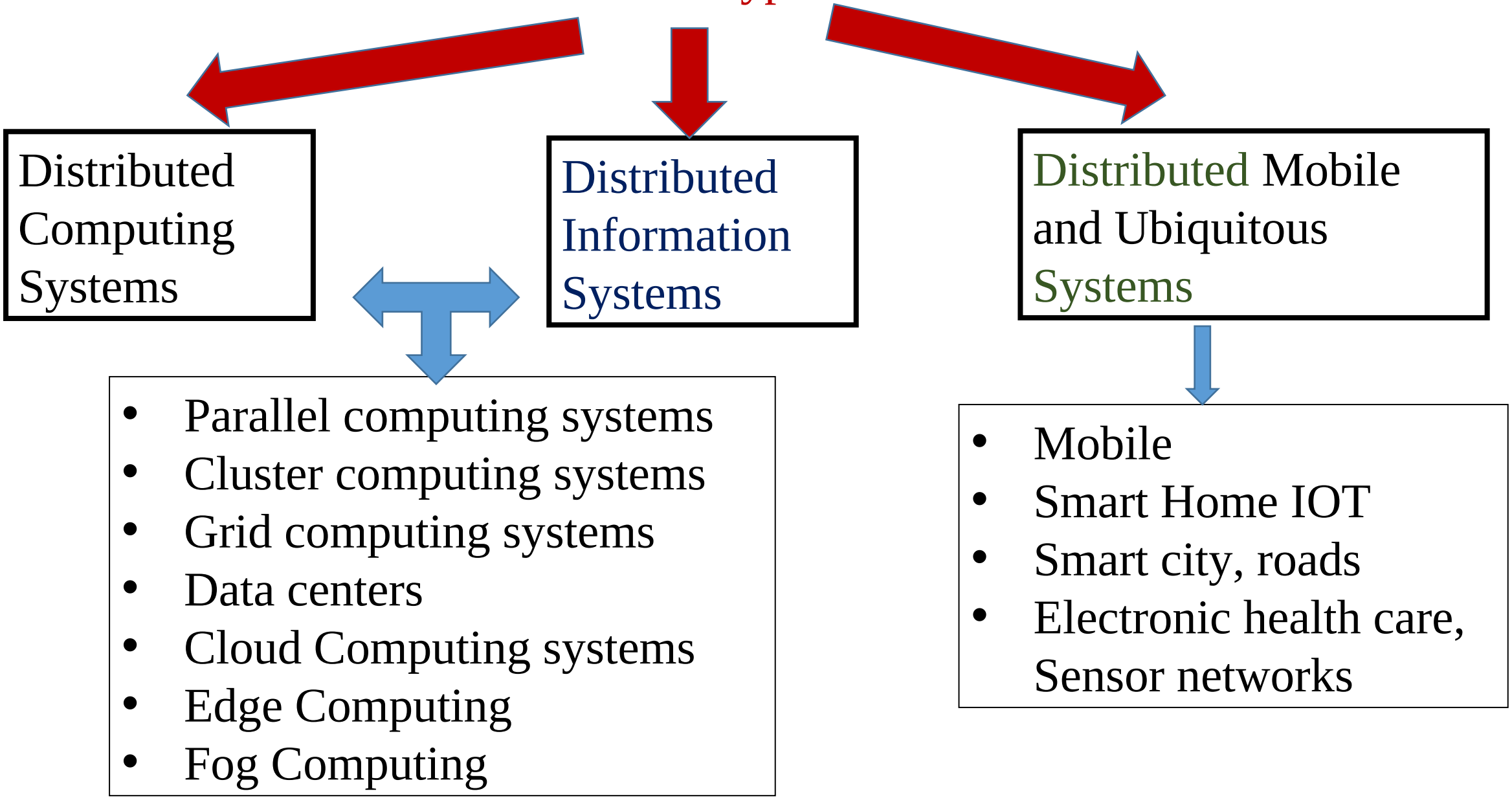
- Multiple autonomous components
- Components are not shared by all users
- Resources accessible on demand
- Software runs in concurrent processes on different processors
- Multiple Points of control
- Multiple Points of failure/solved by clustering

Distributed Computer System DCS Definition

- A DCS is one in which **components** located at networked computers communicate and **coordinate** their actions only by passing **messages**.
- It requires autonomous computers to be interconnected through a network.
- Each computer has to be equipped with DOS software, which enables the computers to coordinate activities and to share resources in a controlled way
- DCS requires transparency of distribution for the computer users. Users shall not have to be aware of the fact that the system is distributed.

- This definition leads to the following characteristics of distributed systems:
- Concurrency of components
- Lack of a global ‘clock’
- Independent failures of components

DCS Types


Distributed Computing Systems
Distributed Information Systems
Distributed Mobile and Ubiquitous Systems
• Parallel computing systems
• Cluster computing systems
• Grid computing systems
• Data centers
• Cloud Computing systems
• Edge Computing
• Fog Computing
• Mobile
• Smart Home IOT
• Smart city, roads
• Electronic health care, Sensor networks

Distributed Mobile and Ubiquitous Systems

- **Distributed Mobile and Ubiquitous Systems DMS:** Distributed Embedded Systems, Home systems, Health care systems, Sensor networks

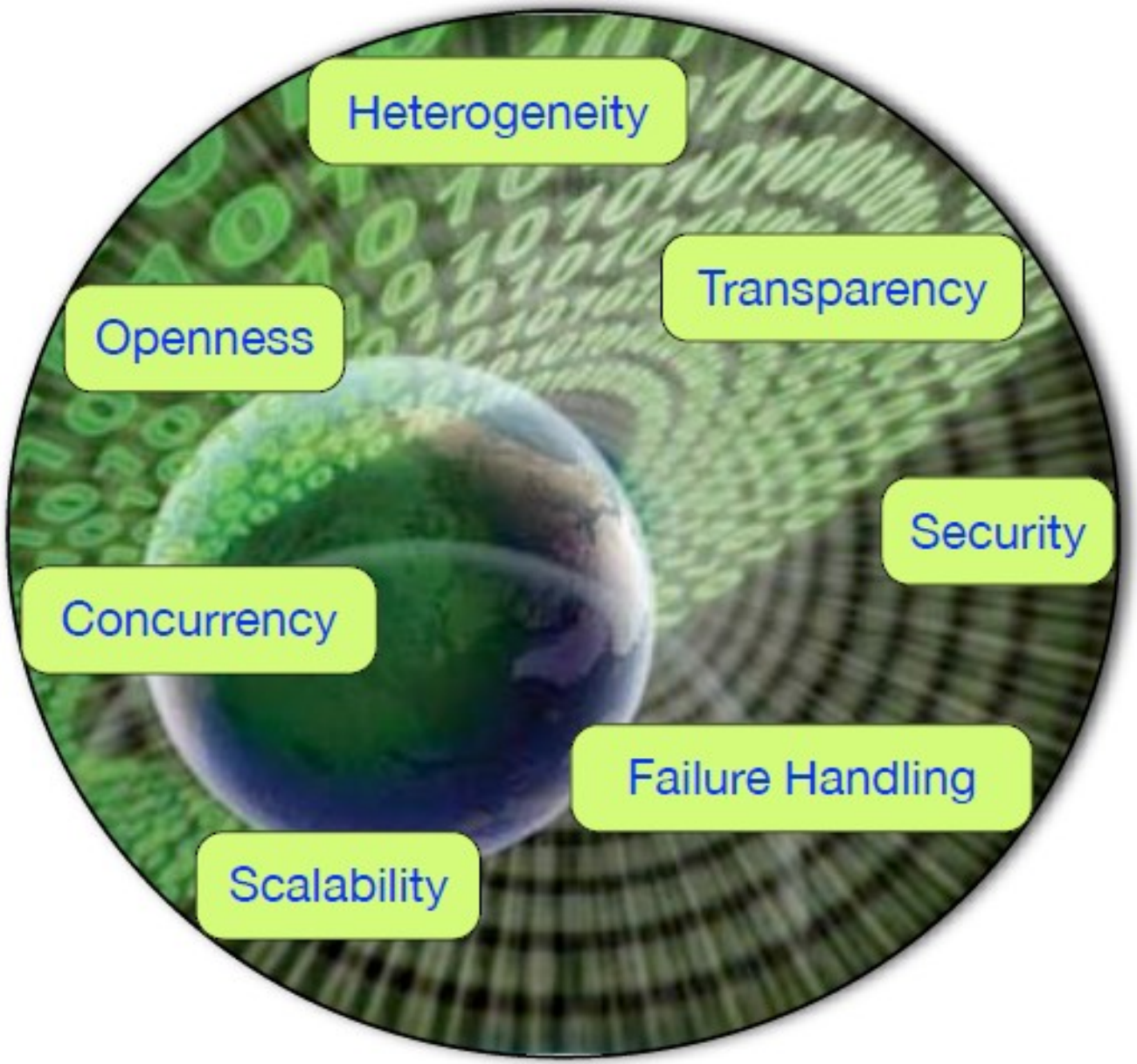


DCS Examples: Intranet and LAN, Internet/WWW, Mobile and Ubiquitous, Database Management System, Automatic Teller Machine Network, E-mail, FTP, Electronic payment, real-time processing.

DCS Characteristics/Design Issues

DCS common characteristics:

1. **Heterogeneity**
2. **Openness**
3. **Concurrency**
4. **Scalability**
5. **Fault tolerance**
6. **Security**
7. **Transparency**

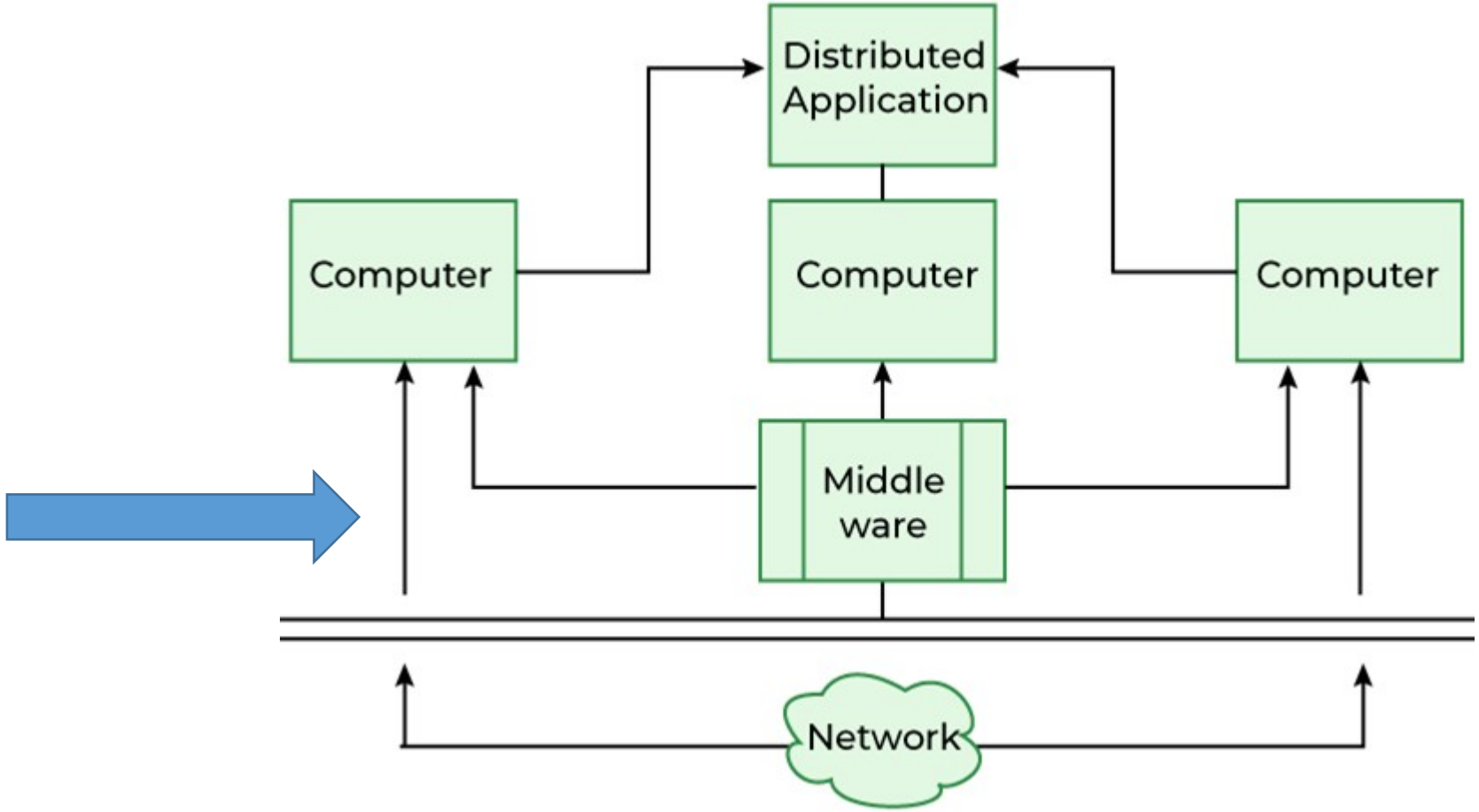


1- Heterogeneity

Heterogeneity is applied to the network, computer hardware, operating system, and implementation/development of different developers.

A key component of the heterogeneous distributed system client-server environment **is middleware operated as Application Program Interface API**

Middleware is a set of services that enables applications and end-user to interact with each other across a heterogeneous distributed system by masking the heterogeneity of the networks, hardware, OS, and programming languages



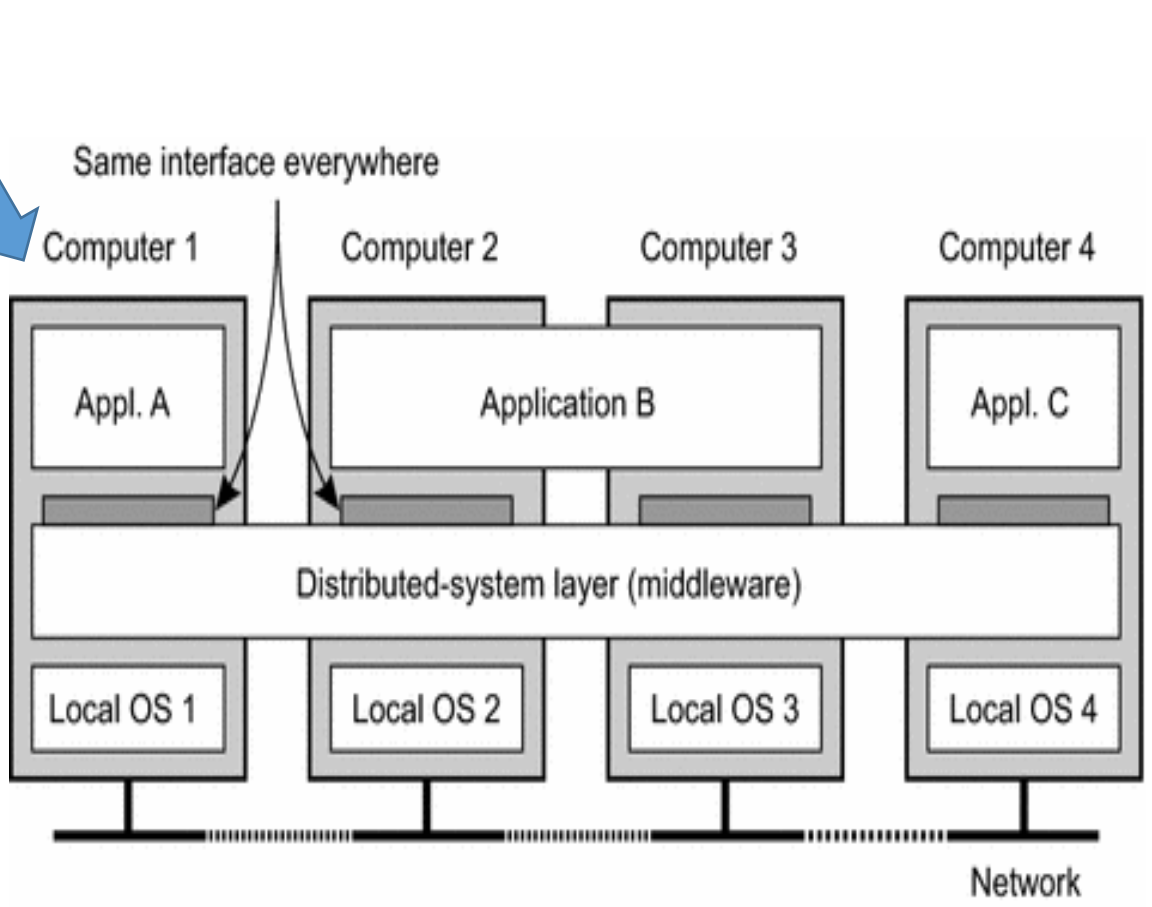
1- Heterogeneity

- Middleware provides a uniform computational model for use by the programmers of servers and distributed applications to communicate between homogenous and heterogenous systems.

- A distributed system organized as middleware.

- The middleware layer extends over multiple machines, and offers each application the same interface

- Possible models include Distributed Transaction Processing Systems DTPS.



2- Openness

- The openness of the DCS is determined by the degree to which new resource-sharing services can be made available to the users.
- Open systems are characterized by the fact that their key interfaces are published.
- It is based on a uniform communication mechanism and published interface for access to shared resources.
- It can be constructed from heterogeneous hardware and software.
- Openness is concerned with extensions and improvements of DCSs.
- Detailed interfaces of components need to be published.
- New components have to be integrated with existing components.
- Differences in data representation of interface types on different processors (of different vendors) have to be resolved.

3- Security

The security of a DCS has three OBJECTIVES: Confidentially, Integrity, and Availability CIA. Encryption protects shared resources and keeps sensitive information secrets when transmitted.

The security of a DS can be achieved by securing the processes and the channels used for their interactions and by protecting the objects that they encapsulate against unauthorized access

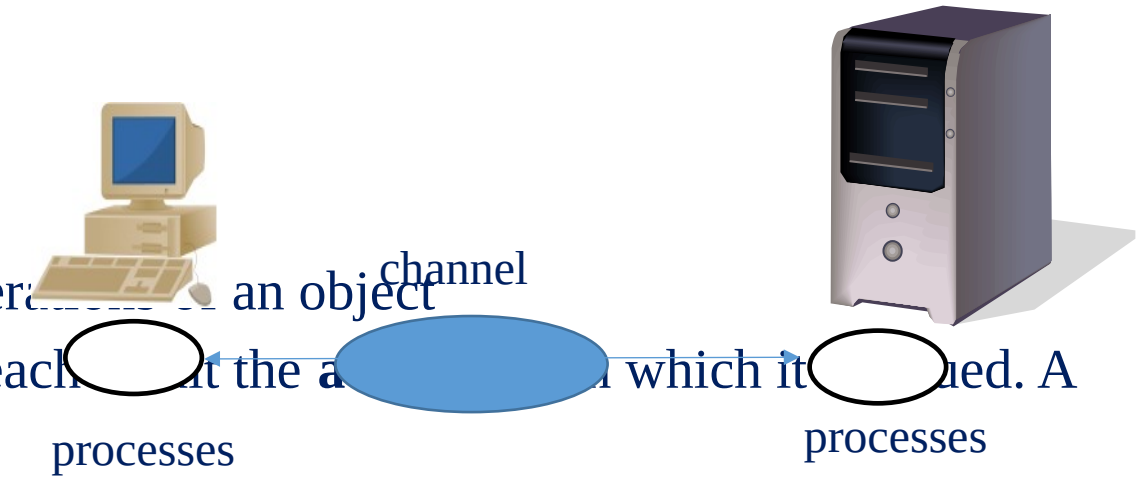
Security Threats include:

1) Threats to the processes
2) Threats to the servers
3) Threats to the communication channel
4) Denial of service DOS attacks

Examples of Security Tools:

1.Access rights – specify who is allowed to perform the operations of an object

2.Principal – associated with each session/association and each object the **a** which it **ued**. A principal may be a user or a process



3.Encryption and authentication- are used to build secure channels as a service layer on top of existing communication services.

4- Scalability

- The scalability of the system should remain efficient even with a significant increase in the number of users and resources connected.
- It shouldn’t matter if a program has 10 or 100 nodes; performance shouldn’t vary.
- A DCS’s scaling requires consideration of a number of elements, including size, geography, and management.
- Adaptation of distributed systems to
  - accommodate more users
  - respond faster (this is the hard one)
- Usually done by adding more and/or faster processors.
- Components should not need to be changed when scale of a system increases.
- Design components to be scalable!

5- Failure Handling (Fault Tolerance)

- When some faults occur in hardware and the software program, it may produce incorrect results or they may stop before they have completed the intended computation so corrective measures should to implemented to handle this case.
- Failure handling is difficult in DCS because the failure is partial i.e., some components fail while others continue to function.
- Hardware, software and networks fail!

- Distributed systems must maintain *availability* even at low levels of hardware/software/network *reliability*.

- Fault tolerance is achieved by
  - recovery
  - redundancy

6- Concurrency

- There is a possibility that several clients will attempt to access a shared resource at the same time.
- Multiple users make requests on the same resources, i.e. read, write, and update.
- Each resource must be safe in a concurrent environment.
- Any object that represents a shared resource in a DCS must ensure that it operates correctly in a concurrent environment.
- Components in distributed systems are executed in concurrent processes.
- Components access and update shared resources (e.g. variables, databases, device drivers).
- Integrity of the system may be violated if concurrent updates are not coordinated.
  - Lost updates
  - Inconsistent analysis

7- Transparency

- Transparency ensures that the DCS should be perceived as a single entity by the users or the application programmers rather than a collection of autonomous systems, which is cooperating.
- The user should be unaware of where the services are located and the transfer from a local machine to a remote one should be transparent.
- Distributed systems should be perceived by users and application programmers as a whole rather than as a collection of cooperating components.

- Transparency has different types.

- These represent various properties that distributed systems should have.

Types of transparency

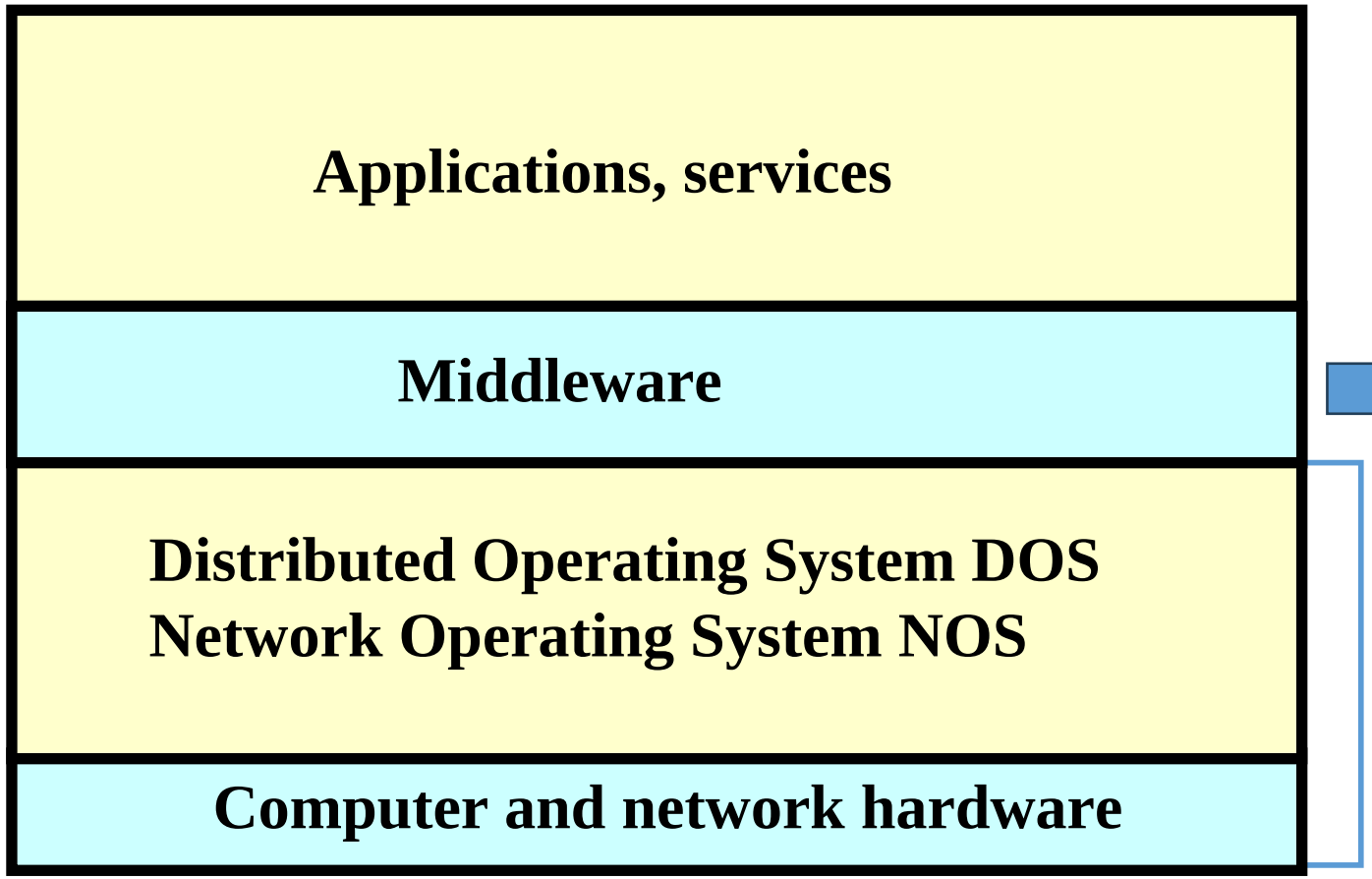
Transparency	Description
Access	Hide differences in data representation and how a resource is accessed
Location	Hide where a resource is located, users cannot tell where hardware and software resources such as Servers, printers, files, data bases are located.
Migration	Hide that a resource may move to another location
Relocation	Hide that a resource may be moved to another location while in use
Replication	Hide that a resource is replicated, OS can make additional copies of files and resources without users noticing.
Concurrency	Hide that a resource may be shared by several competitive users, The users are not aware of the existence of other users. Need to allow multiple users to concurrently access the same resource. Lock and unlock for mutual exclusion.
Failure	Hide the failure and recovery of a resource
Concurrency Transparency	Automatic use of parallelism without having to program explicitly. The objectives of distributed and parallel system designers

Distributed Computer Systems DCS Technologies

- DCS is a collection of **autonomous computer systems** that are physically separated but are connected by a **computer network** that is equipped with **distributed system software**.
- DCS refer to a network of computers that work together to achieve a common goal.
- In a DCS, individual computers are connected to each other through a communication network, and they coordinate their activities by exchanging messages.

  - DCSs can be used for a variety of purposes, including **data centers, cloud computing, online gaming, web applications, distributed database, distributed multimedia, processing large amounts of data, running complex simulations, and providing high-performance computing resources for scientific research.**
  - DCS combine the computational power of multiple computers systems **(hardware+ software)** connected via a **network** to solve complex problems/applications.
  - The individual computers in a DCS are typically spread over wide geographies (LAN and or WAN), and possess heterogeneous processors, OS, and DBMS architectures.

Software Layers in DCS (Software Architecture)



To mask the heterogeneity & to provide a convenient Programming model

Platform

Ex. x86/Windows, SPARC/Solaris, PowerPC/MacOS, x86/Linux, novel NetWare, IBM AX,…

Difference between Network OS NOS and Distributed OS DOS

NOS

- A NOS is an operating system that is expected to support networking functionalities.
- This enables to connect several computers and to share many resources, e.g. files, applications, and printers.
- Although each computer works self-contained it can communicate with other users in the network through the NOS.

NOS Examples: MS SERVER, IBM AIX, ORCLE SOLARES, LINUX SERVER, UNIX

Advantages of Network OS:

- **Resource Sharing**
- **Centralized Control**
- **Scalability**
- **NOS Security**

DOS

- DOS is responsible for coordinating the computers/nodes that are connected and operates in cooperation like a single large machine.
- In DOS every computer operates as an individual OS, a DOS unifies all computers and they work together.

DOS EXAMPLES: MS AZURE, GOOGLE FILE SYSTEM GFS, LOCUS, ORCLE SOLARES

Advantages of Distributed OS:

- **Resource Utilization**
- **Transparency**
- **Fault Tolerance**
- **Scalability**
- **DOS Security**

Distributed Computer Systems DCS Technologies

1.Middleware

Middleware is a set of services that enables applications and end-user to interact with each other across a heterogeneous DCS by **masking the heterogeneity of the underlying networks, hardware, OS, and programming languages**

2.Computing Cluster

A Computing Cluster is a set of connected computers that work together so closely that in many respects they function as a single computer.

The idea is to take a large computational task, break it down into discrete parts, and distribute those parts across different computers to be processed simultaneously.

This leads to a significant increase in processing speed and efficiency, allowing tasks to be completed more quickly and effectively than with a single computer.

3.Virtualization

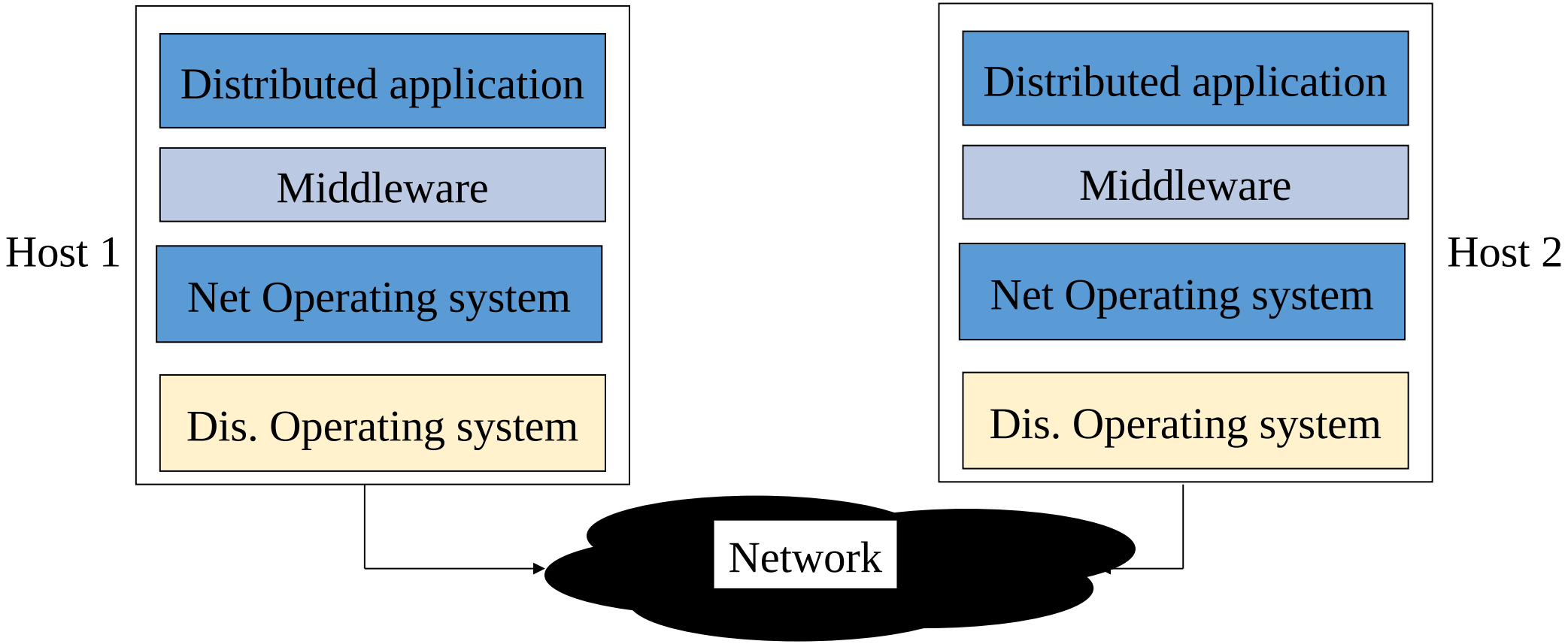
Virtualization is the ability to run multiple OSs on a physical system and share the hardware, software, storage, Desktop, printing, and network resources

Virtualization is a technique, which allows to share single physical instance of an application or resource among multiple organizations or customers.

It does so by assigning a logical name to a physical resource and providing a pointer to that physical resource when demanded

1. Middleware

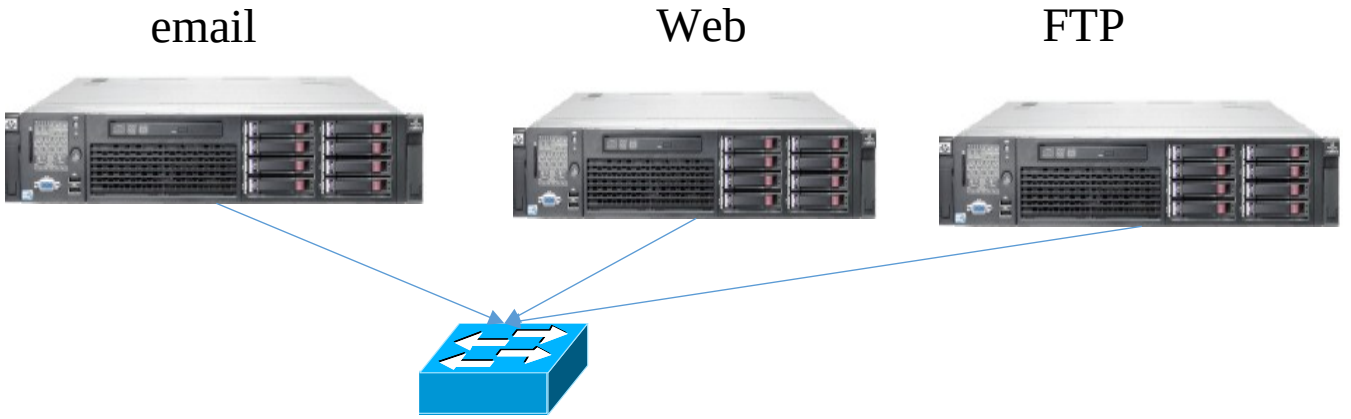
- Middleware is Layer between (DOS and NOS) and distributed applications
- Hides complexity and heterogeneity of distributed system
- Bridges gap between low-level OS communications and programming language abstractions
- Provides common programming abstraction and infrastructure for distributed applications



Clustering Distributed Computing Virtualization

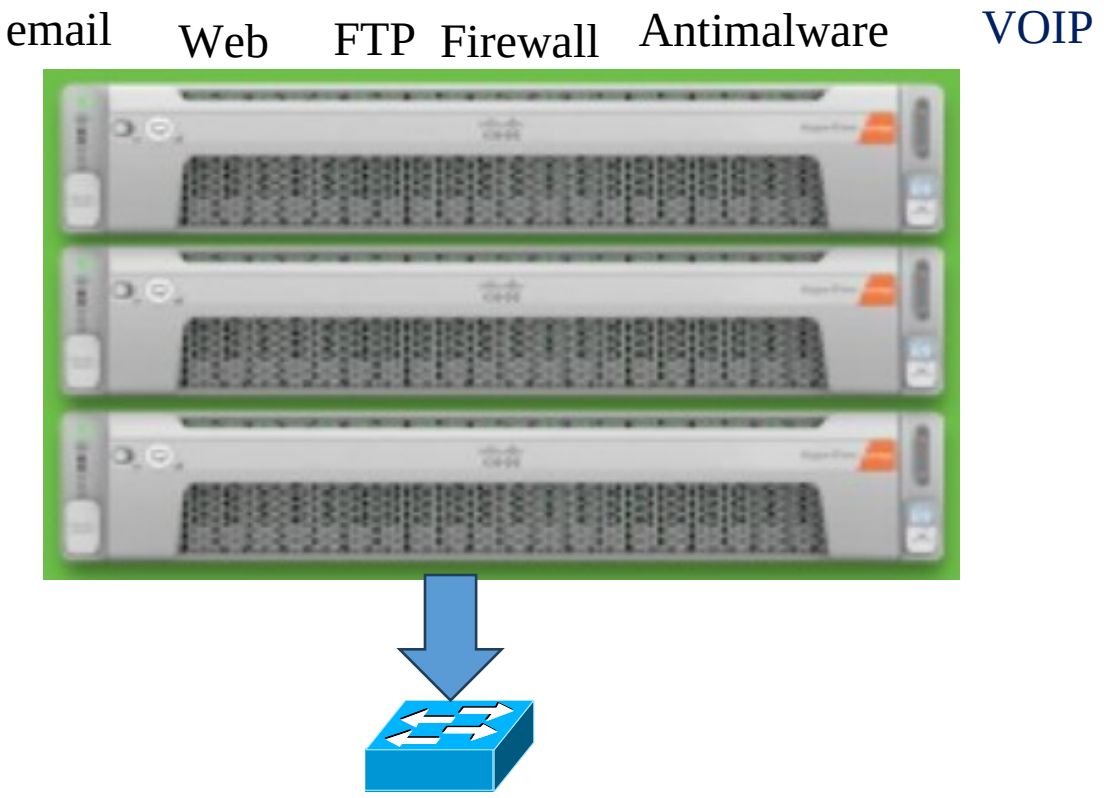
Traditional System

Servers run email 30% utilization, 40% Web, and 30%FTP



Virtualization based system

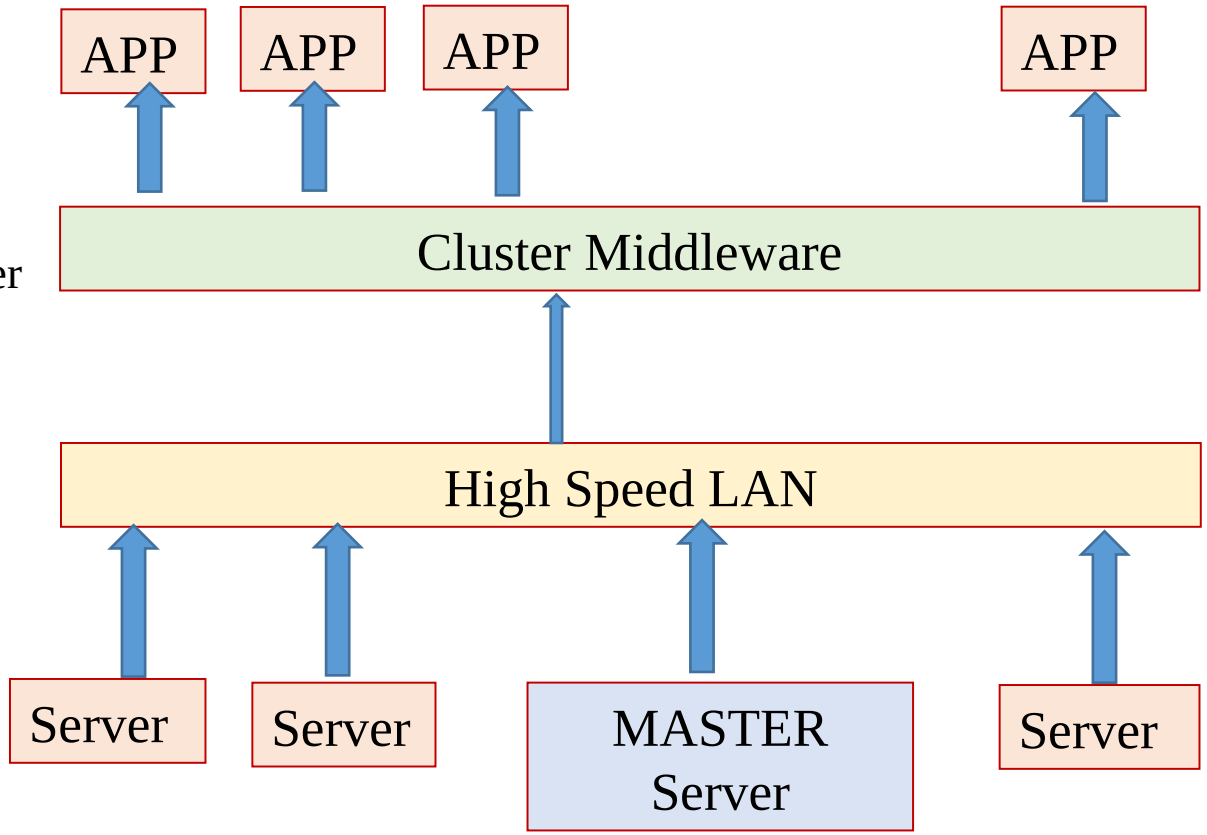
Servers Cluster with 3 nodes run email, Web, FTP, Firewall, security antimalware And VOIP 60% utilization



- HCI Nodes All-Flash HX240C.
  - 16 CPU
  - 512 cores, 1024 threats.
  - 3 TB Memory
  - 200 TB SSD Storage

2. Cluster Computing

- A cluster is a group of linked computers.
- Working together closely so that in many respects they form a single computer
- The components of a cluster are connected to each other through fast LANs.
- Clusters are deployed to improve performance and/or availability over that provided by a single computer, while being much more cost effective than single computers of comparable speed or availability



Cluster consists of:

Nodes (master + computing)

Network

Oss

Cluster middleware which permits compute clustering programs to be portable to a wide variety of clusters

Cluster Classification

High availability Clusters (HA) (Linux)

- Mission critical applications
- High availability /Failover clusters are implemented for the purpose of improving the availability of services which the cluster provides
- Provides redundancy
- Eliminate single point of failure

Network Load balancing Clusters

- Operate by distributing a workload over multiple backend nodes
- The cluster will be configured with multiple redundant load-balancing front ends
- All available servers processes (Web servers, mail servers, ftp servers,….) requests

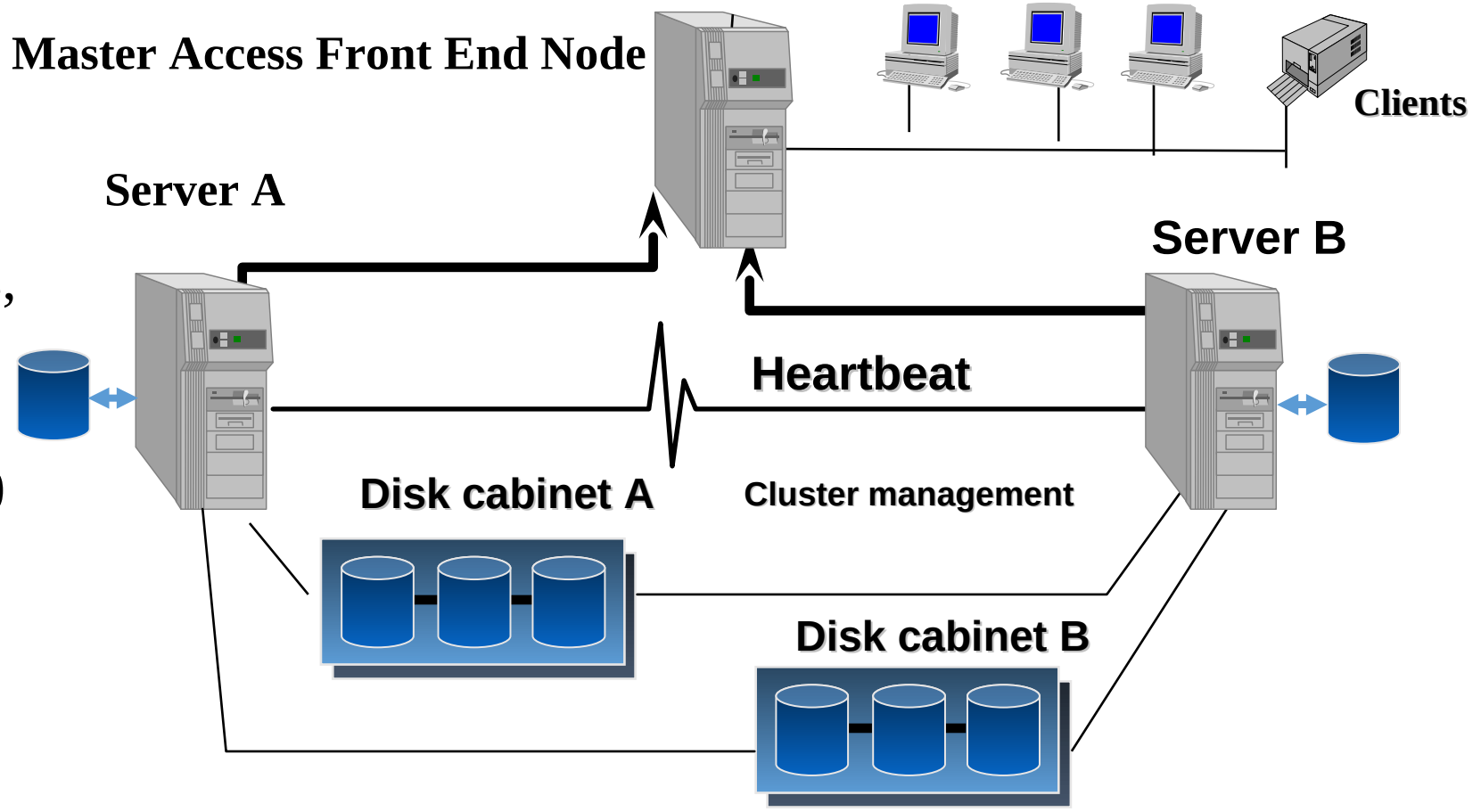
Science Clusters

- Science cluster is a computer design that uses parallel and cluster processing across multiple computers to create cheap powerful supercomputer
- It has two types of computers, a master/front/service computer and a node computers
- When a large problem or set of data is given, the master computer first runs a program that breaks the problem into discrete pieces, it then sends a piece to each node to compute
- As nodes finish their tasks, the master computer sends more piece to them until the entire problem has been computed

Clustering Distributed Computer Systems DCS

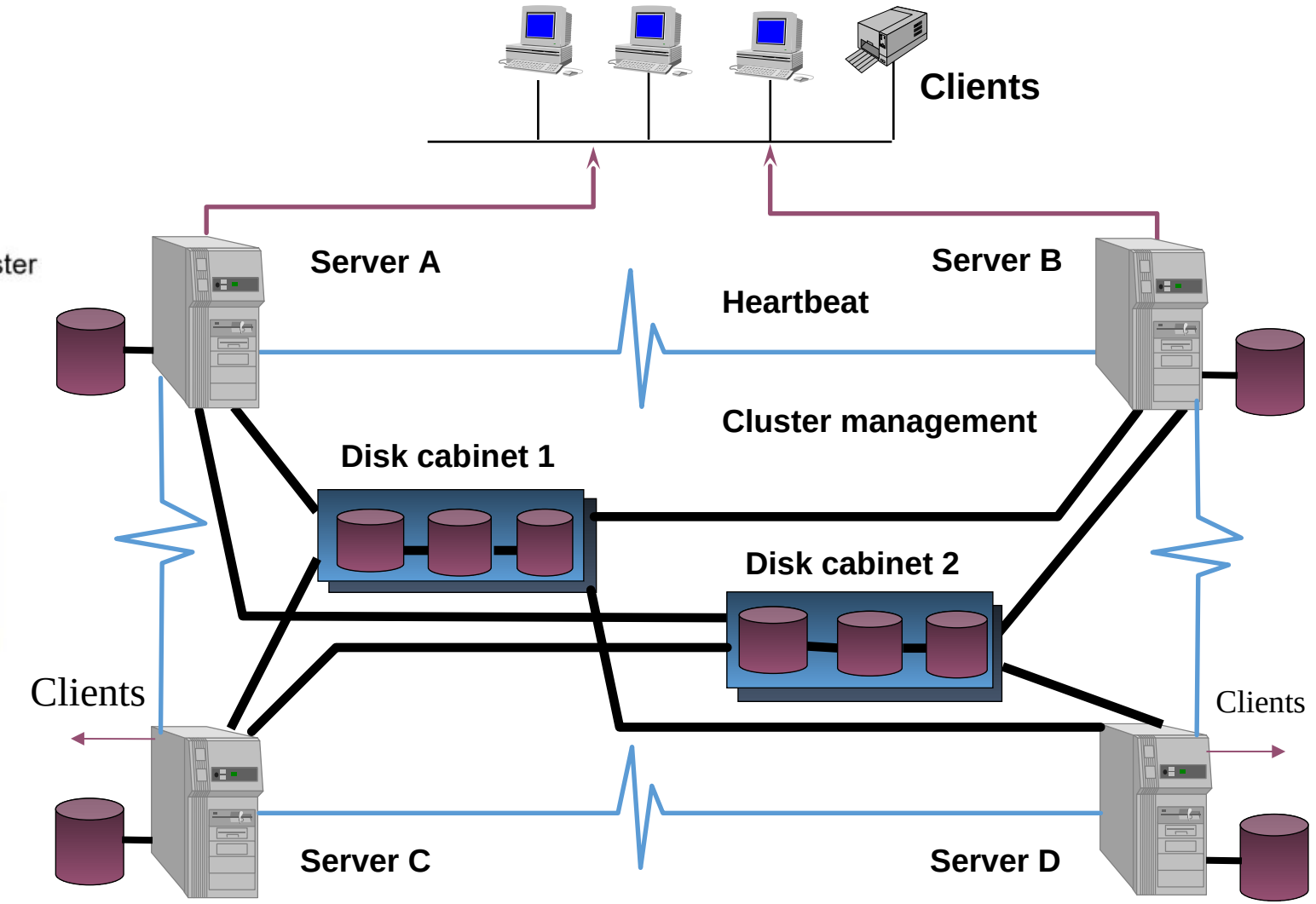
- A **computer cluster** is a collection of interconnected stand-alone computers which can work together collectively and cooperatively as a single **integrated computing resource pool**.
- **Computer cluster c**onsists of a collection of interconnected whole computers, used as a single, unified computing resource

Cluster Components:
1.Cluster Hardware
2.Communication (Interconnects, Communication Protocols)
3.System Middleware (management, availability, tools)
4.High-performance IO systems (storage, file systems, data placement and movement)



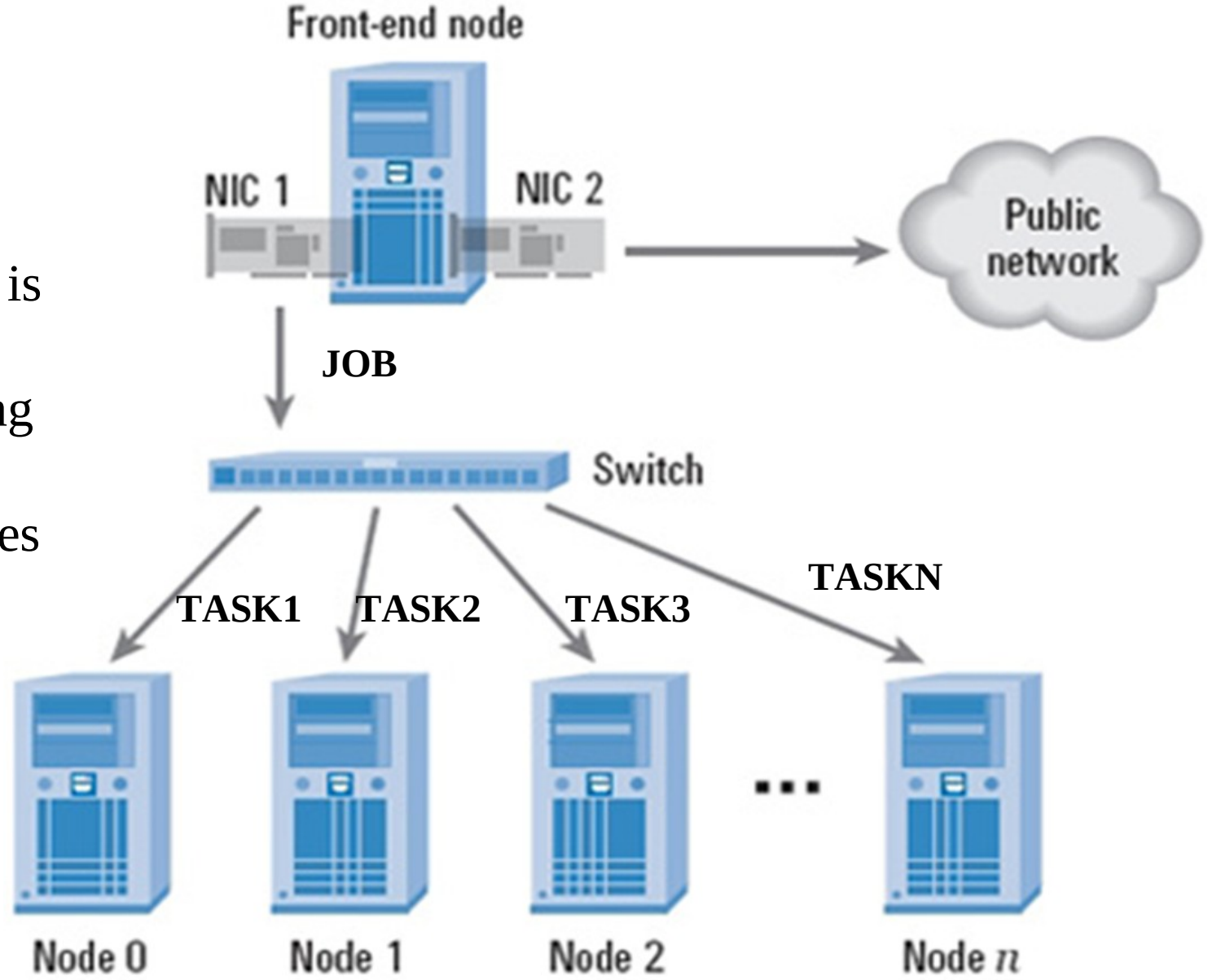
Typical Clustering Distributed Computer Systems DCS





2. Cluster Computing

- Clustering of computers enables scalable parallel and distributed computing in **both science and business applications**.
- The overall goal of cluster computers is to provide a range of cloud services, including high-performance computing capabilities, mass storage (e.g. data centers), and richer application services such as web search.



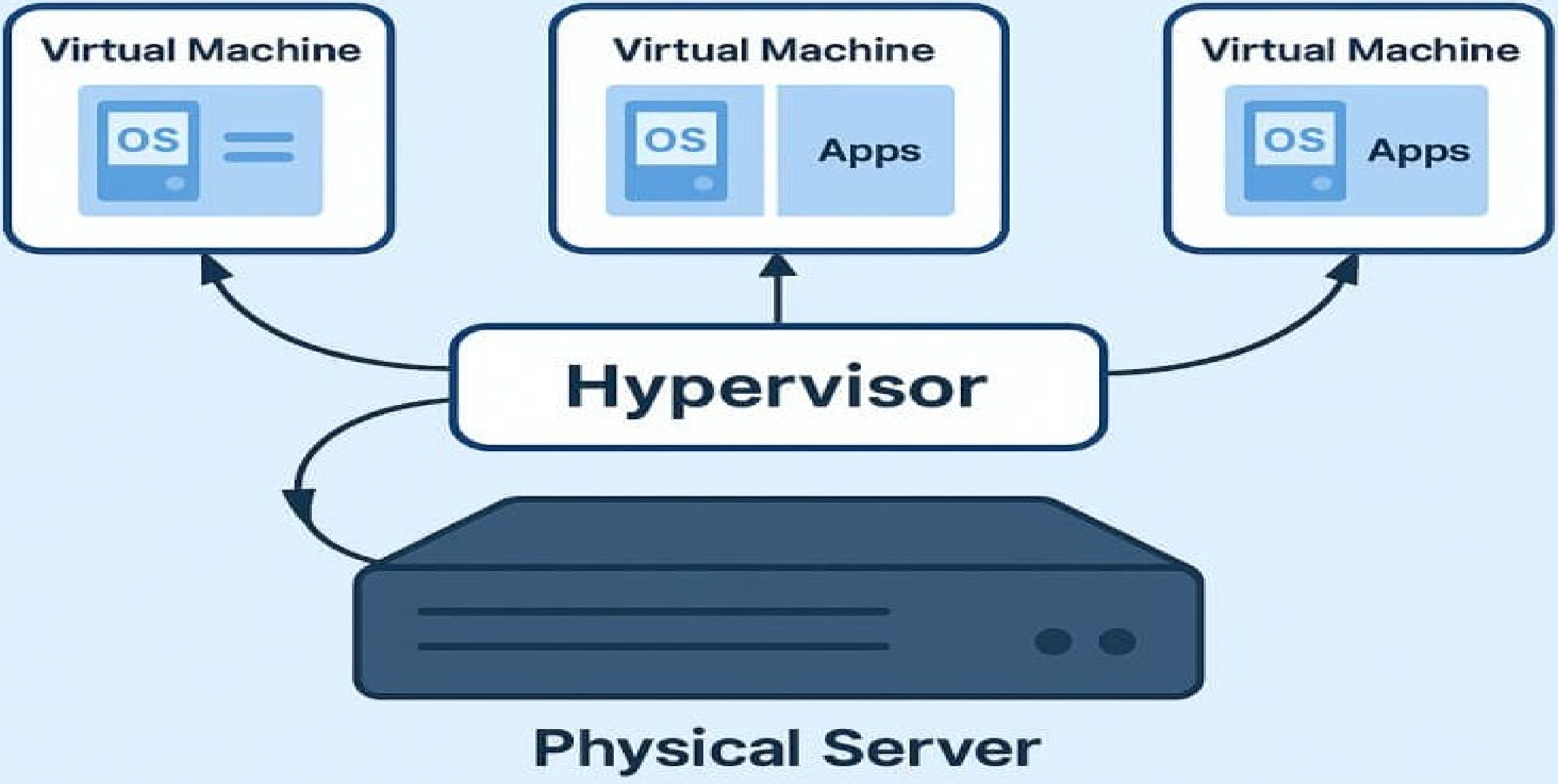
3. Virtualization

- Virtualization is a technique of partitioning the resources of a single computing platform into multiple segregated, virtualized, execution environments.
  - Each environment runs independently of the other, thus allowing multiple OSs to run on the same hardware.
    - The concept of virtualization already present in every-day computing…
      - Most modern OSs contain a simplified system of virtualization;
    - Each running process is able to act as if it is the only thing running.
      - The CPUs and memory are virtualized.
- CPU Virtualization: If a process tries to consume all of the CPU, the OS will preempt it and allow other processes to execute; If an OS tries to consume all of the CPU, the hypervisor will preempt it and allow other processes to execute;
  - Memory virtualization: a running process has its own virtual address space that the OS maps to physical memory to give the process the illusion that it is the only user of RAM.

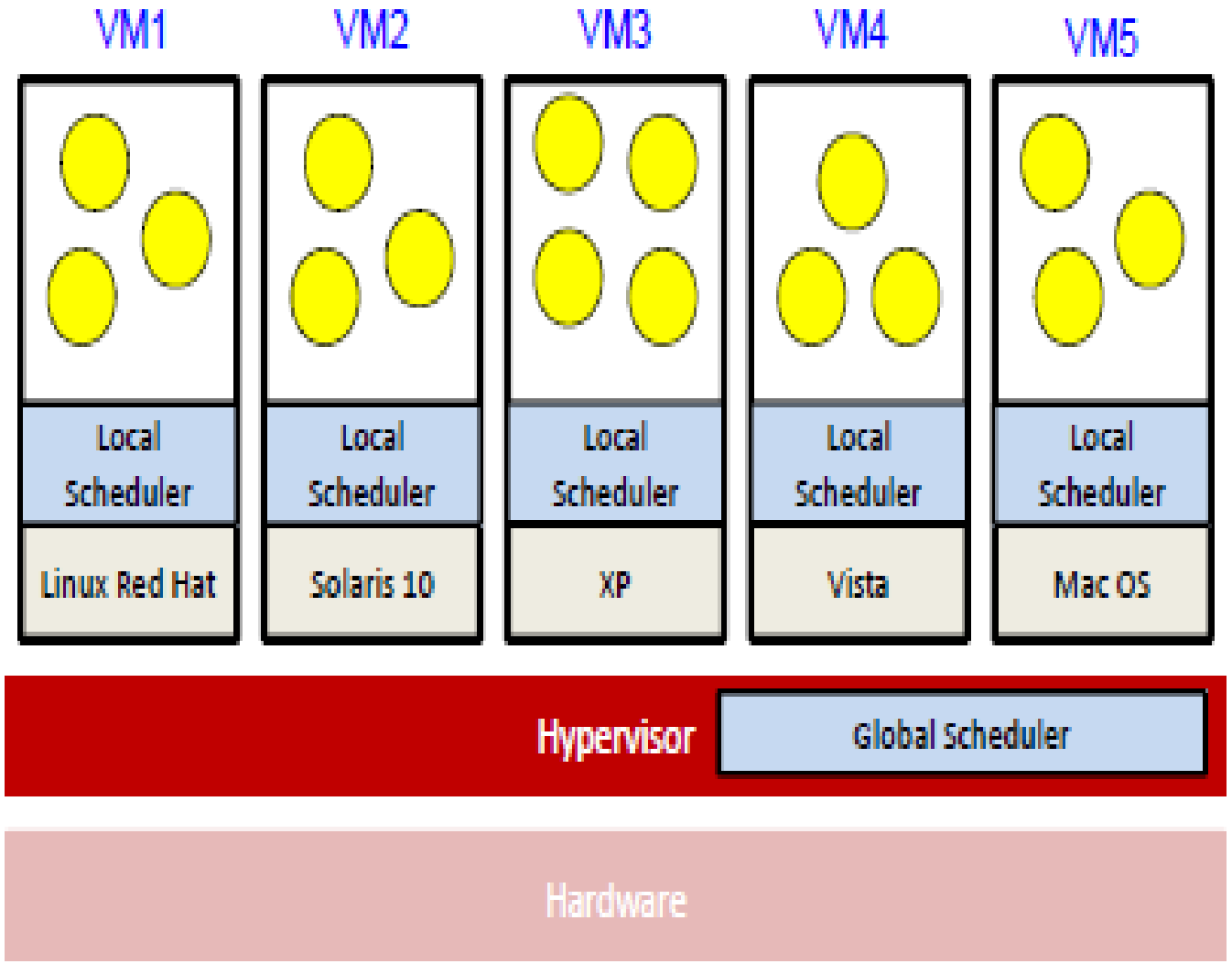
3. Virtualization

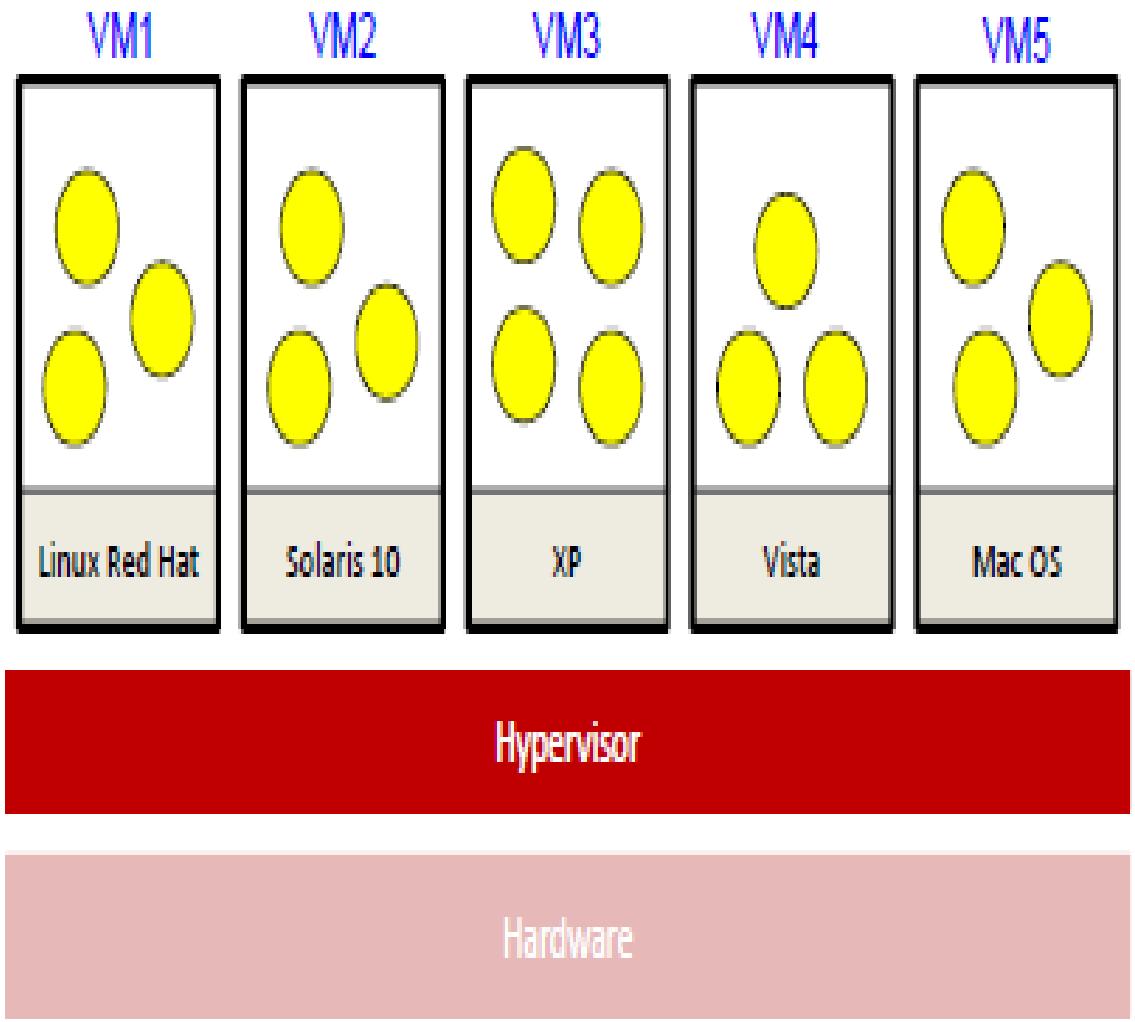
- A running OS has its own virtual address space that the hypervisor maps to physical memory to give the process the illusion that it is the only user of RAM.
  - Each execution environment is called a *guest* and the computing platform on which they execute is called the *host*.
  - The software enabling these multiple execution environments is commonly referred to as **Hypervisor** or Virtual Machine Monitor (VMM).
    - The Hypervisor runs on the host and acts as a *bridge* between the host and the guests;
  - Multiple VMs can be implemented on a single hardware platform to provide individuals or user groups with their own OS environments

3. Virtualization



3. Virtualization

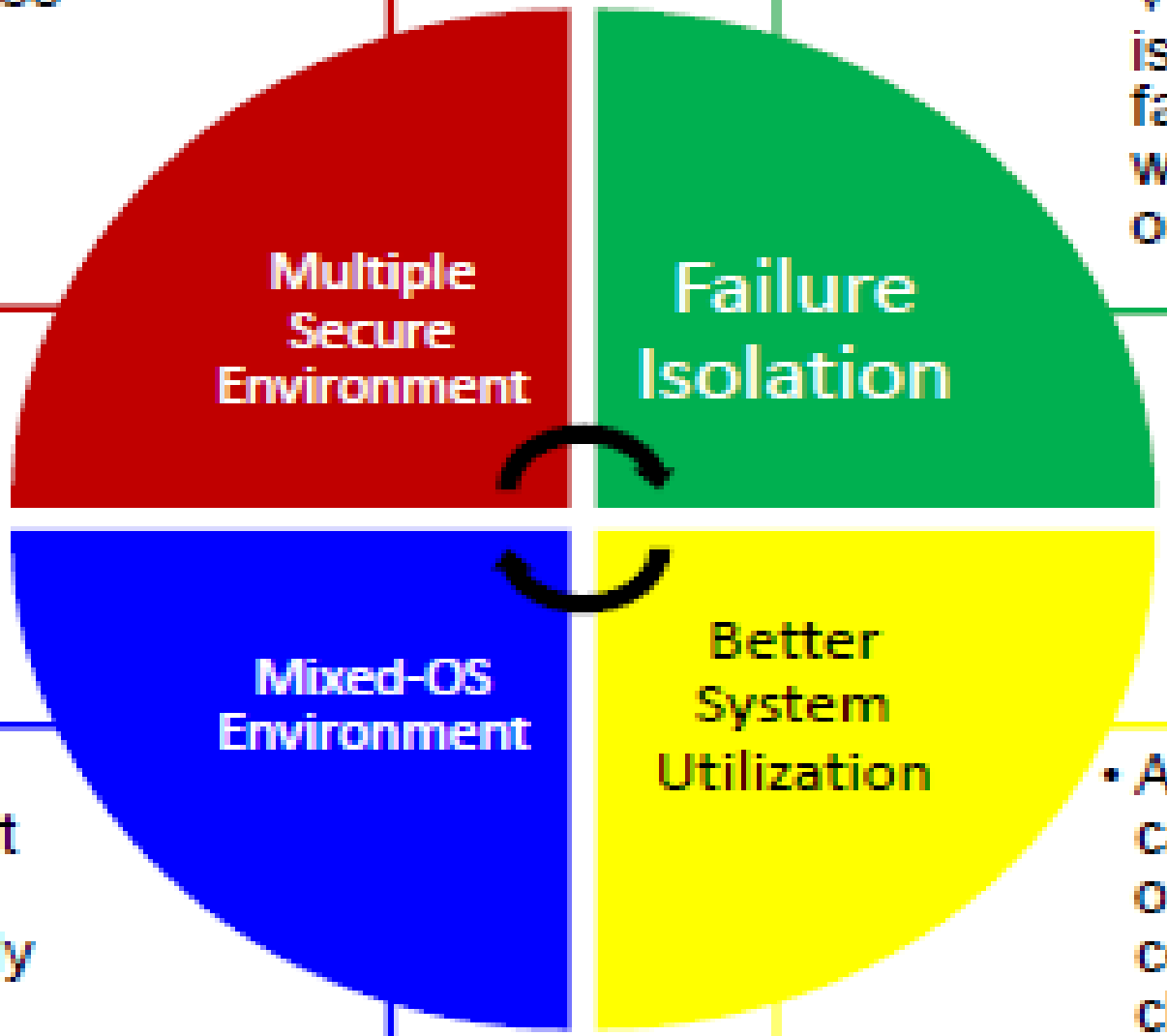




Virtualization Advantages

- A system VM provides a sandbox that isolates one system environment from other environments

- Virtualization helps isolate the effects of a failure to the VM where the failure occurred



- A single hardware platform can support multiple operating systems concurrently

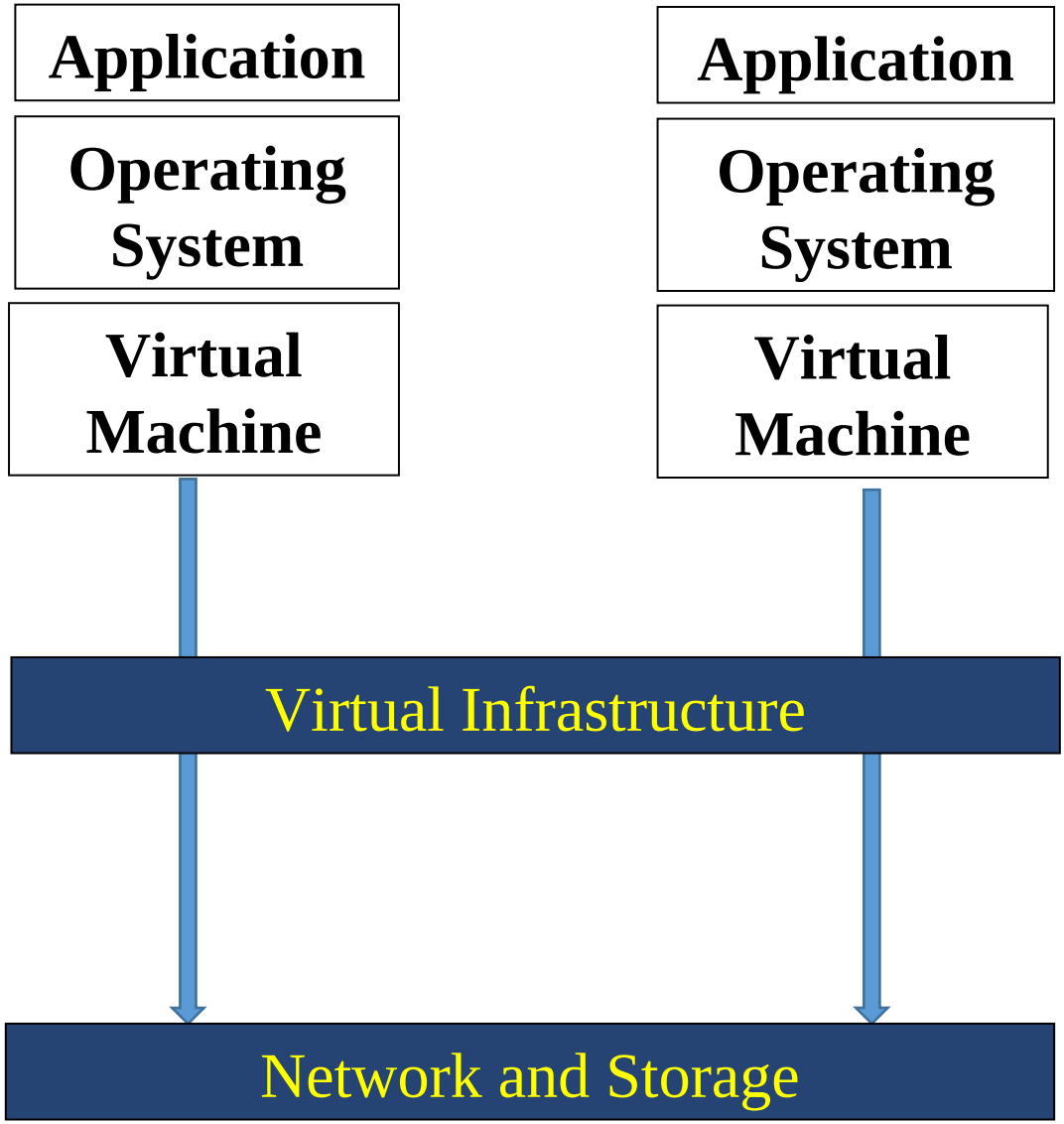
- A virtualized system can be (dynamically or statically) re-configured for changing needs

Advantages of Virtualization

- **Multiple secure environment:** a system VM provides a sandbox that isolates one system environment from other environments
- **Mixed OS environment:** a single hardware platform can support multiple Oss concurrently
- **Better system utilization:** a virtualized system can be (dynamically or statically) reconfigured for changing needs
- **Failure isolation:** virtualization helps isolate the effects of a failure to the VM where the failure occurred

Virtualization Techniques

- Virtualization is a technique, which allows to share single physical instance of an application or resource among multiple organizations or customers.
- It does so by assigning a logical name to a physical resource and providing a pointer to that physical resource when demanded
- The multitenant architecture offers virtual isolation among the multiple tenants and therefore the organizations can use and customize the application as though they each have its own instance



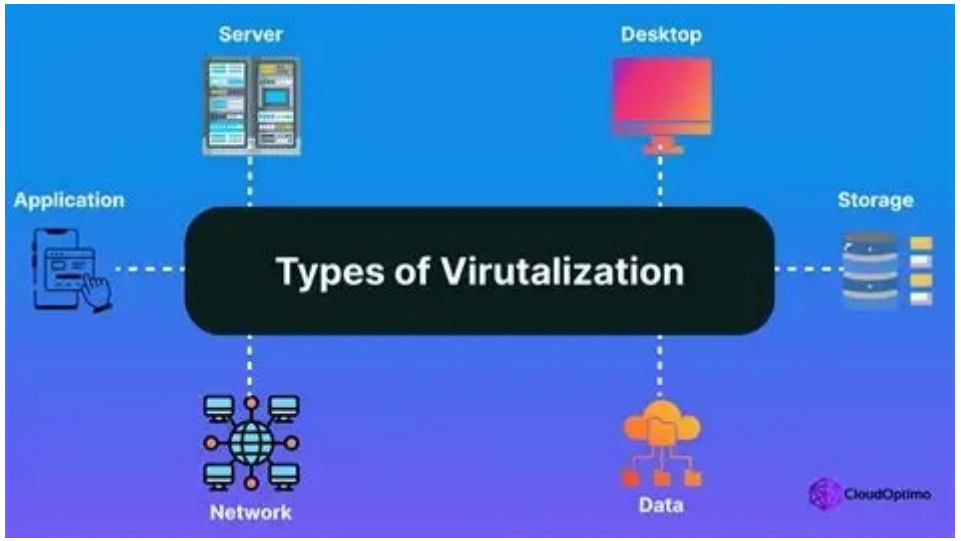
Virtualization Techniques

- Virtualization enables sharing of resources much easily, it helps in increasing the degree of hardware level parallelism, basically, there is sharing of the same hardware unit among different kinds of independent units, if we say that we have the same physical hardware and on that physical hardware, we have multiple OSs.
  - There can be different users running on different kind of OSs. Therefore, we have a much more processing capability with us.
- This also helps in increasing the degree of hardware parallelism as well as there is a reduced hardware footprint throughout the VM consolidation.
  - The hardware footprint that is overall hardware consumption also reduces out the amount of hardware that is wasted out that can also be reduced out.
- This consequently helps in easing out the management process and also to reduce the amount of energy that would have been otherwise consumed out by a particular hardware if we would have invested in large number of hardware machines would have been used otherwise.
  - Virtualization helps in sandboxing capabilities or migrating different kinds of applications that in turn enables flexible allocations and utilization of the resources.

Distributed Computing Virtualization

- **Virtualization** is the ability to run multiple OSs on a physical system and share the hardware, software, storage, Desktop, printing, and network resources

Server Virtualization	Network Virtualization	Storage Virtualization
Application Virtualization	Data Virtualization	Desktop Virtualization

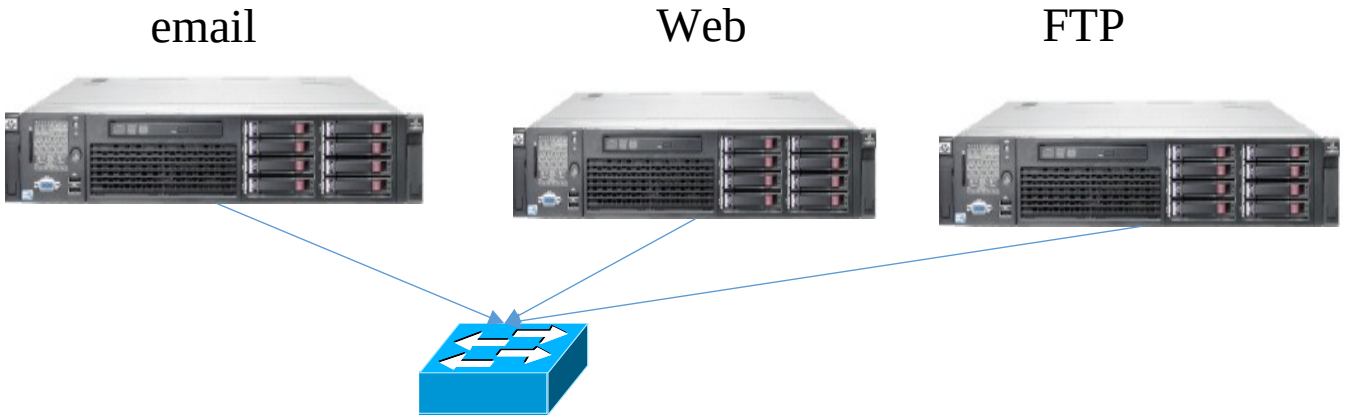


Server Virtualization
Network Virtualization
Storage Virtualization
Application Virtualization
Data Virtualization
Desktop Virtualization

Distributed Computing Virtualization

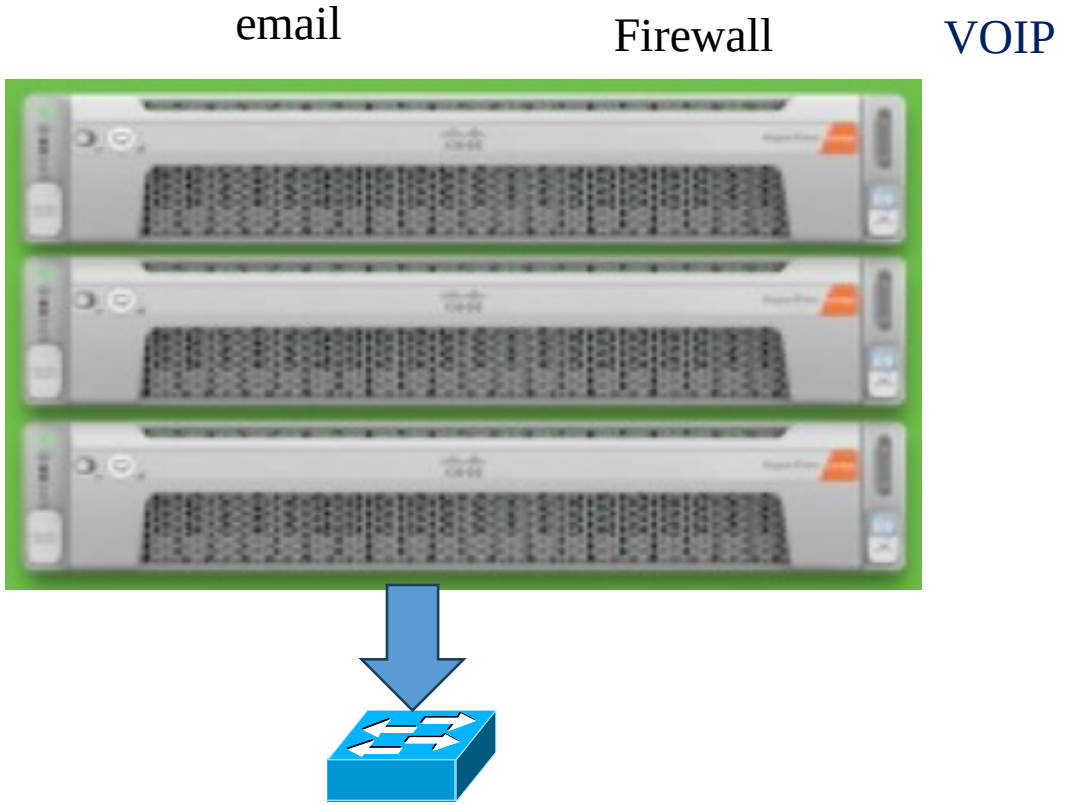
Traditional System

Servers run email 30% utilization, 40% Web, and 30%FTP



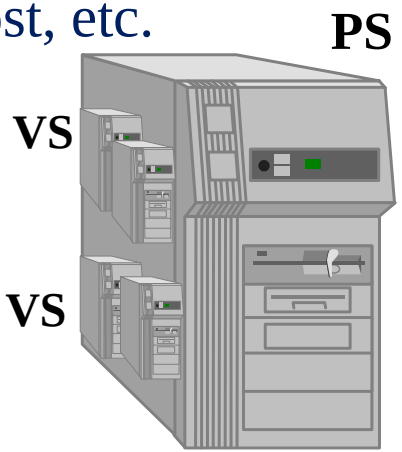
Virtualization based system

Servers Cluster with 3 nodes run email, Web, FTP, Firewall, security antimalware And VOIP 60% utilization

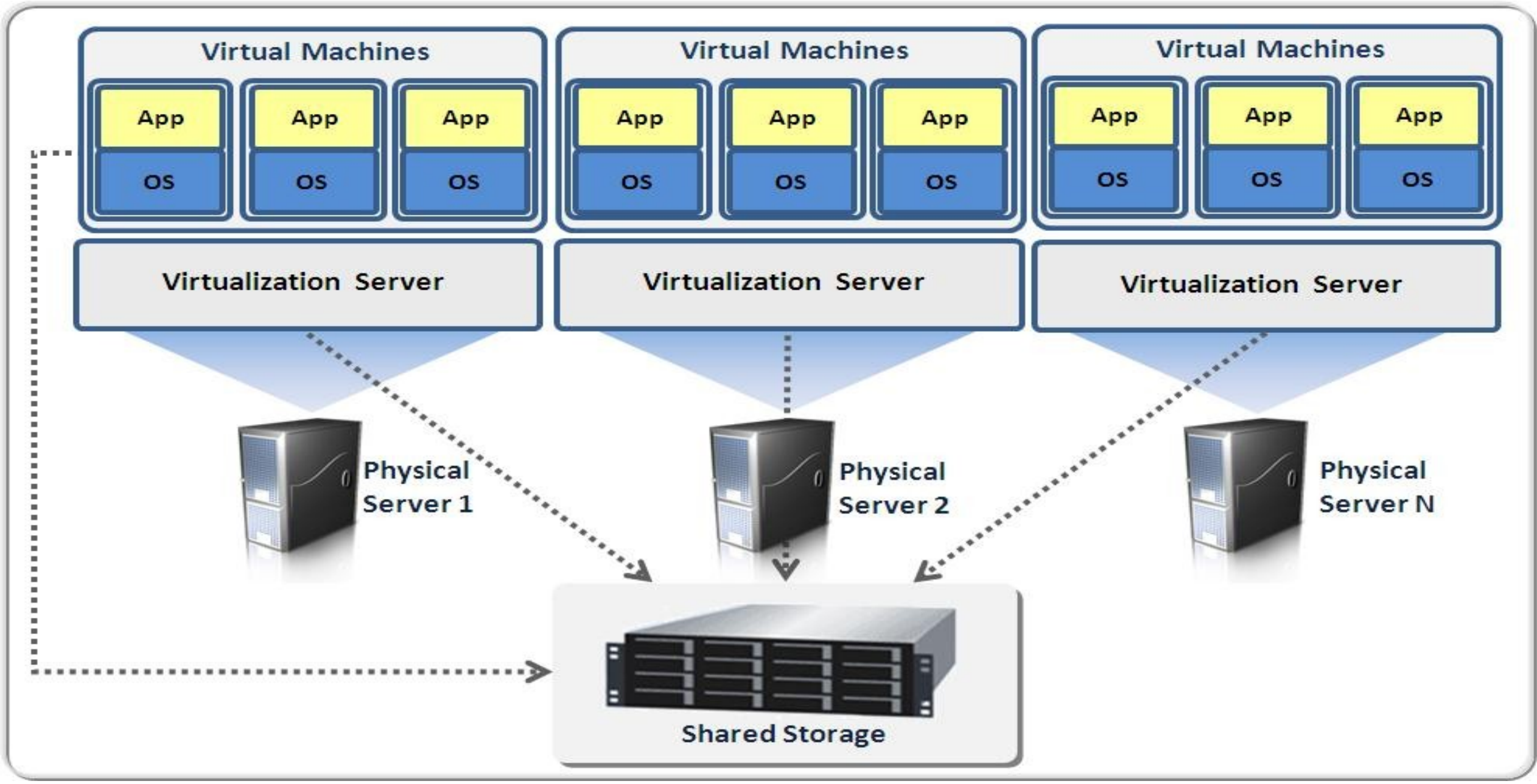


Server Virtualization

- Server virtualization is the process of dividing a Physical Server PS into multiple unique and isolated Virtual Servers VSs by means of a software application.
- Each VS can run its own Server cores and OS independently.
- This is a kind of virtualization in which masking of server resources takes place.
- The central-server (Physical Server PS) is divided into multiple different VSs by changing the identity number and processing cores.
- Each system can operate its own OSs in isolate manner.
- Where each sub-server knows the identity of the central server.
- It causes an increase in the performance and reduces the operating cost by the deployment of main server resources into a sub-server resource.
- It’s beneficial in virtual migration, reduce energy consumption, reduce infrastructural cost, etc.



Server Virtualization

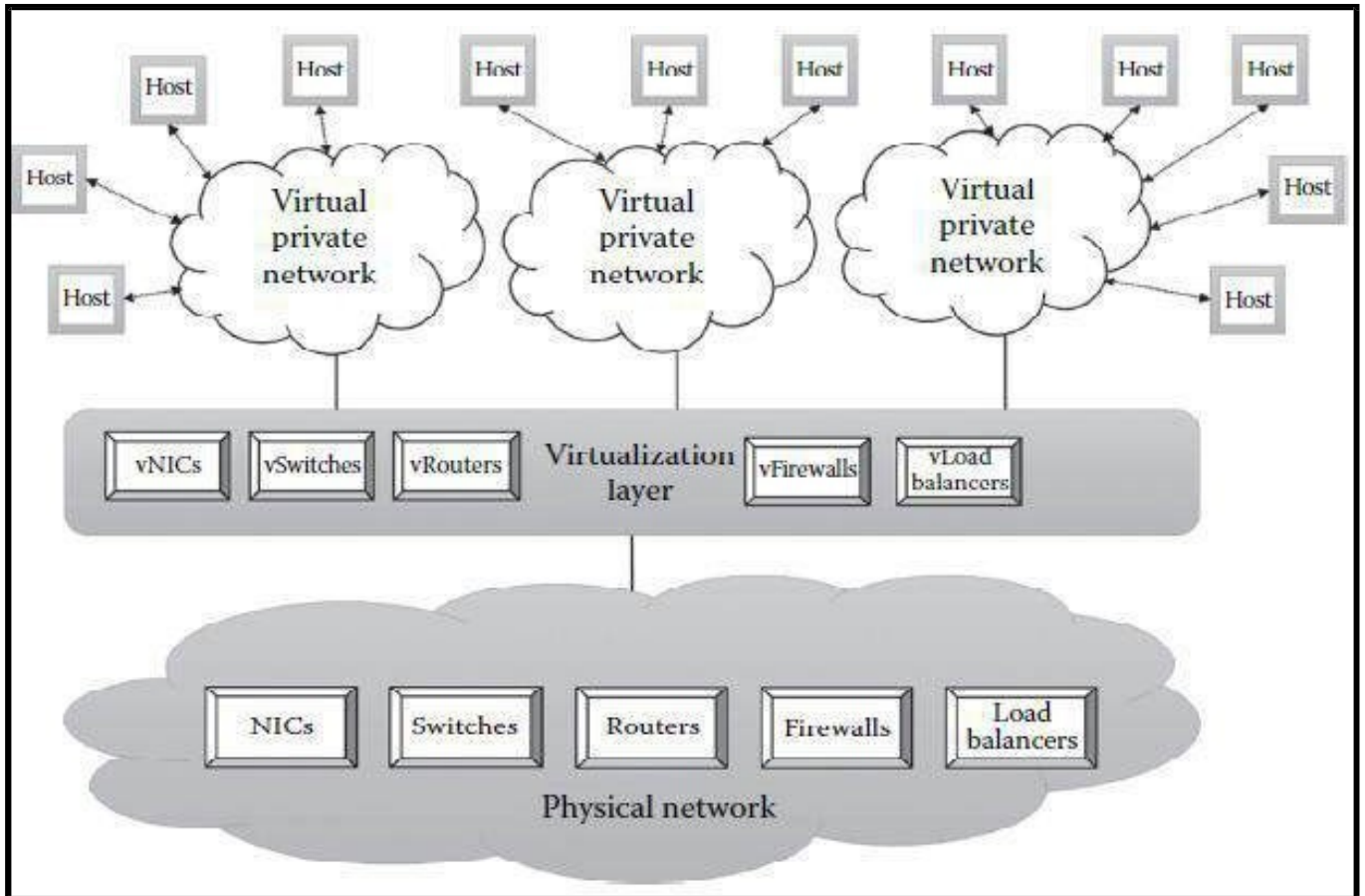


Network Virtualization

- The ability to run multiple virtual networks with each has a separate control and data plan. It co-exists together on top of one physical network. It can be managed by individual parties that potentially confidential to each other.
- Network Virtualization provides a facility to create and provision Virtual Networks VN—logical switches, routers, firewalls, load balancer, Virtual Private Network (VPN), and workload security within days or even in weeks.
- VN is a process of logically grouping physical networks and making them operate as single or multiple independent networks.

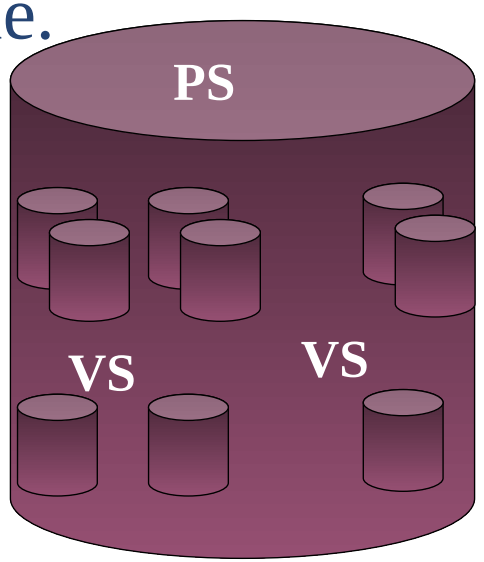
Network Virtualization

- The physical network components like router, switch, and Network Interface Card (NIC) will be controlled by the virtualization software to provide Virtual Network VN components.
  - VN is a single software-based entity that contains the network hardware and software resources.
  - VN can be achieved from internal network or by combining many external networks.
    - It enables the communication between the VNs that share the physical network.



Storage Virtualization

- Storage virtualization is an array of servers that are managed by a Virtual Storage VS system.
- The servers aren’t aware of exactly where their data is stored.
- It makes managing storage from multiple sources to be managed and utilized as a single repository.
- Storage virtualization is the pooling of Physical Storage PS from multiple storage devices into what appears to be a single storage device -- or pool of available storage capacity -- that is managed from a central console.



Application Virtualization

- Application virtualization helps a user to have remote access of an application from a server.
- The server stores all personal information and other characteristics of the application but can still run on a local workstation through the internet.
- Application virtualization software allows users to access and use an application from a separate computer than the one on which the application is installed.
- Using application virtualization software, IT admins can set up remote applications on a server and deliver the apps to an end user’s computer.

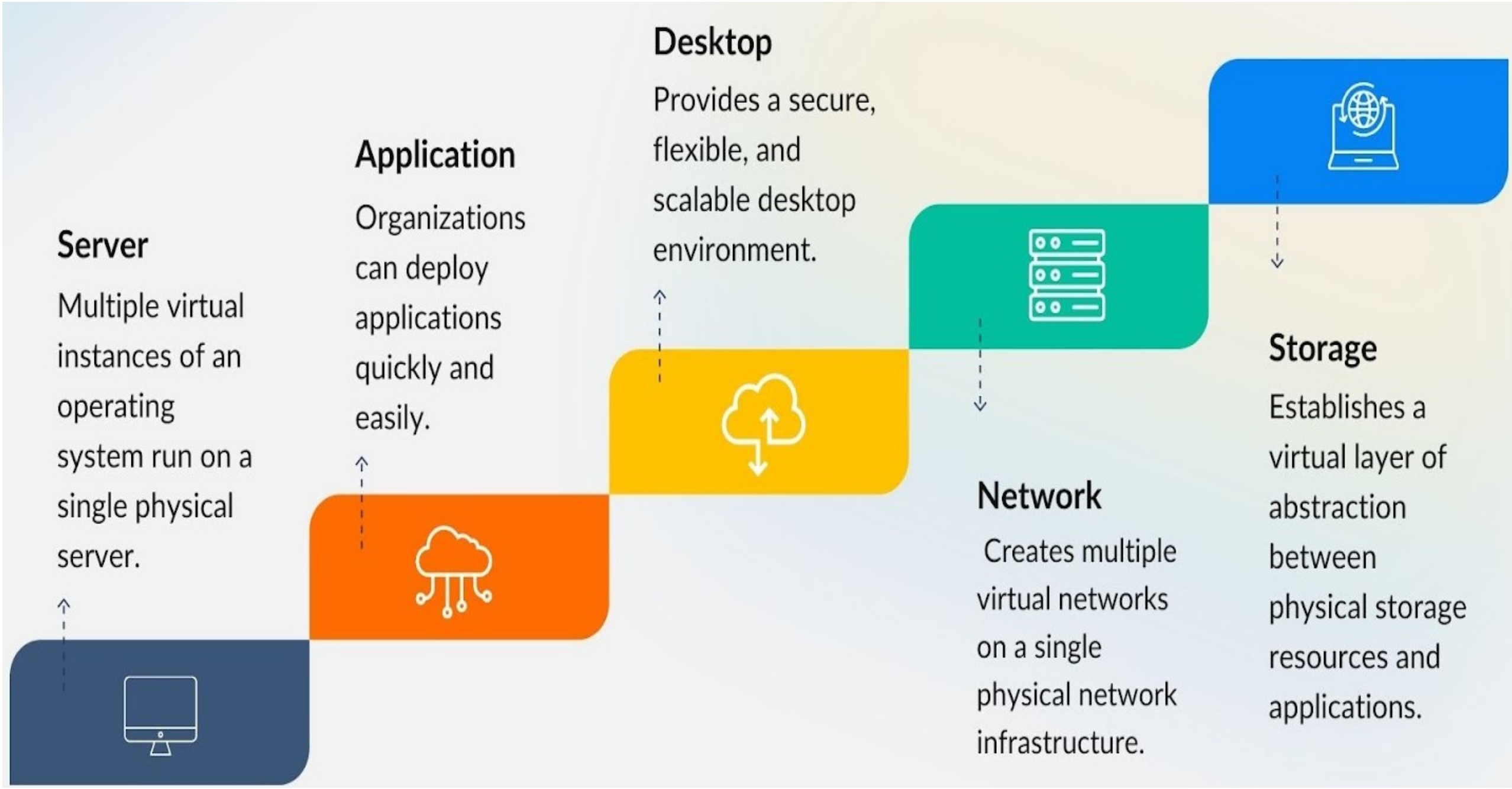
Data Virtualization

- This is the kind of virtualization in which the data is collected from various sources and managed that at a single place without knowing more about the technical information like how data is collected, stored & formatted then arranged that data logically so that its virtual view can be accessed by users through the various cloud services remotely.

Desktop Virtualization

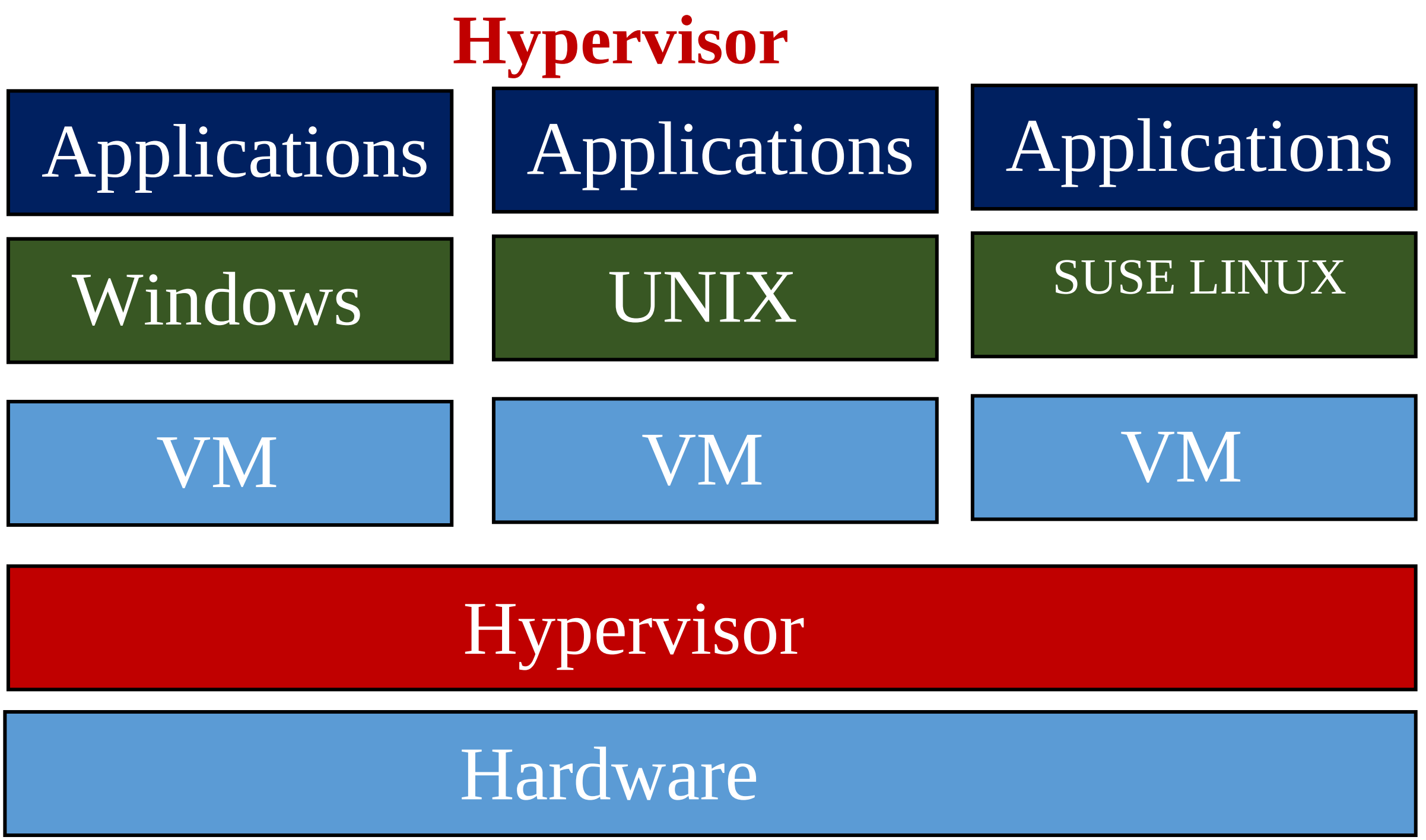
- Desktop virtualization allows the users' OS to be remotely stored on a server in the data center.
- It allows the user to access their desktop virtually, from any location by a different machine.
- Users who want specific operating systems other than Windows Server will need to have a virtual desktop.
- Main benefits of desktop virtualization are user mobility, portability, easy management of software installation, updates, and patches.

Applications and types of Virtualization in Data Center And Cloud Computing



Hypervisor

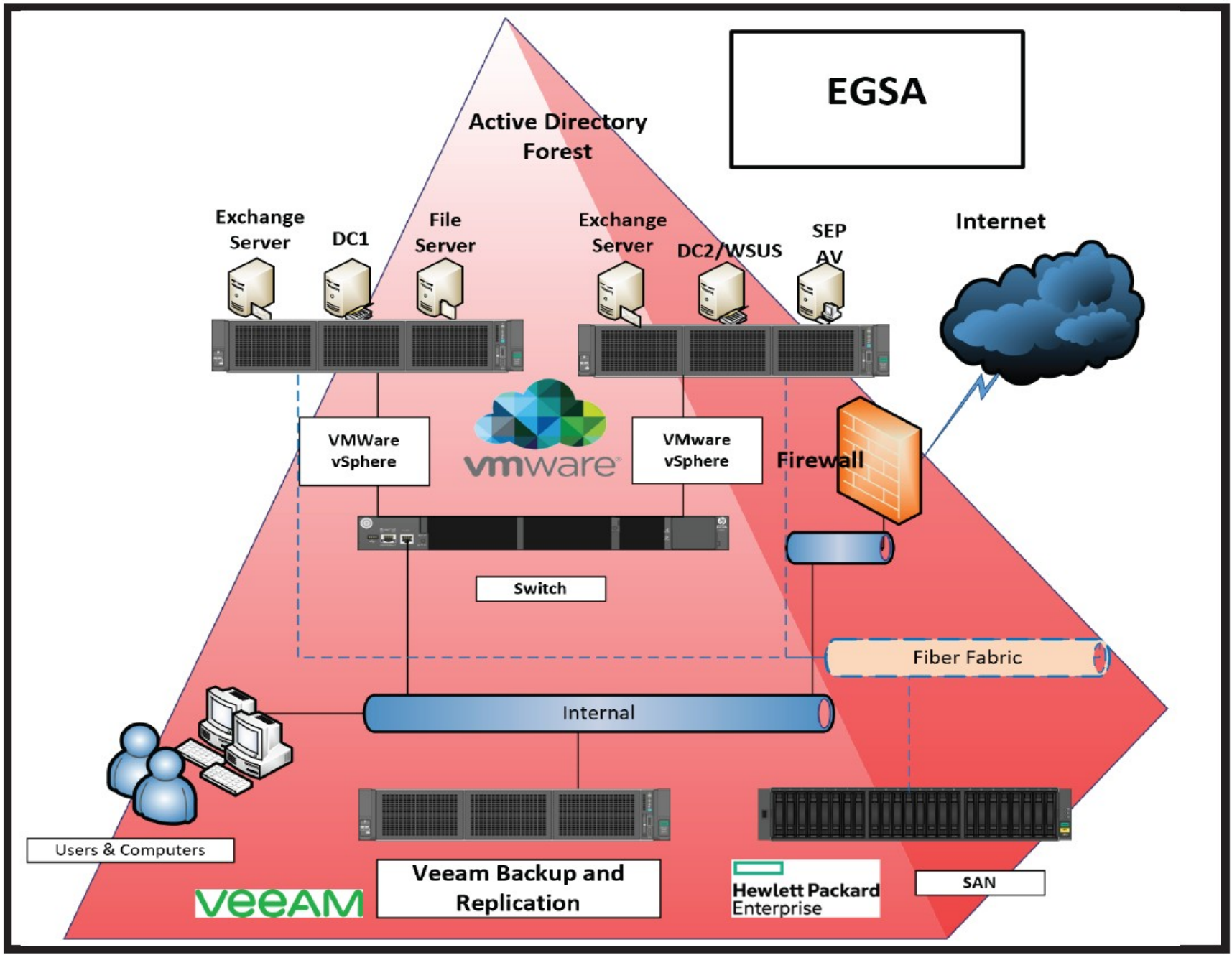
- A hypervisor is a type of software or hardware used to create virtual machines and then run those virtual machines day to day. A hypervisor allows one host computer to support multiple guest VMs by virtually sharing its resources, such as memory and processing.
- A hypervisor/V**irtual Machine Monitor (VMM),** isolates the hypervisor operating system and resources from the virtual machines and enables the creation and management of those VMs.
- The VMM is installed on the top of host OS. The VMM allows the users to run different kinds of guests OSs within their own application window of a particular hypervisor.
- **Different kinds of hypervisors can be Oracle's VirtualBox, Microsoft Virtual PC, VMware Workstation.**


Hypervisor
Applications
Applications
Applications
Windows
UNIX
SUSE LINUX
VM
VM
VM
Hypervisor
Hardware

Ex: E mail system based on virtualization

S/W INFRASTRUCTURE

- OS: MS SERVER
- MS Exchange SERVER
- MS SQL
- VMware vSphere
- Windows Server Update Service (WSUS).
- DC: Domain Controller
- AV: Trend Micro antimalware
- Veeam Desktop



CHAPER 1: Distributed Computer System DCS Definition

1- Autonomous Computer System

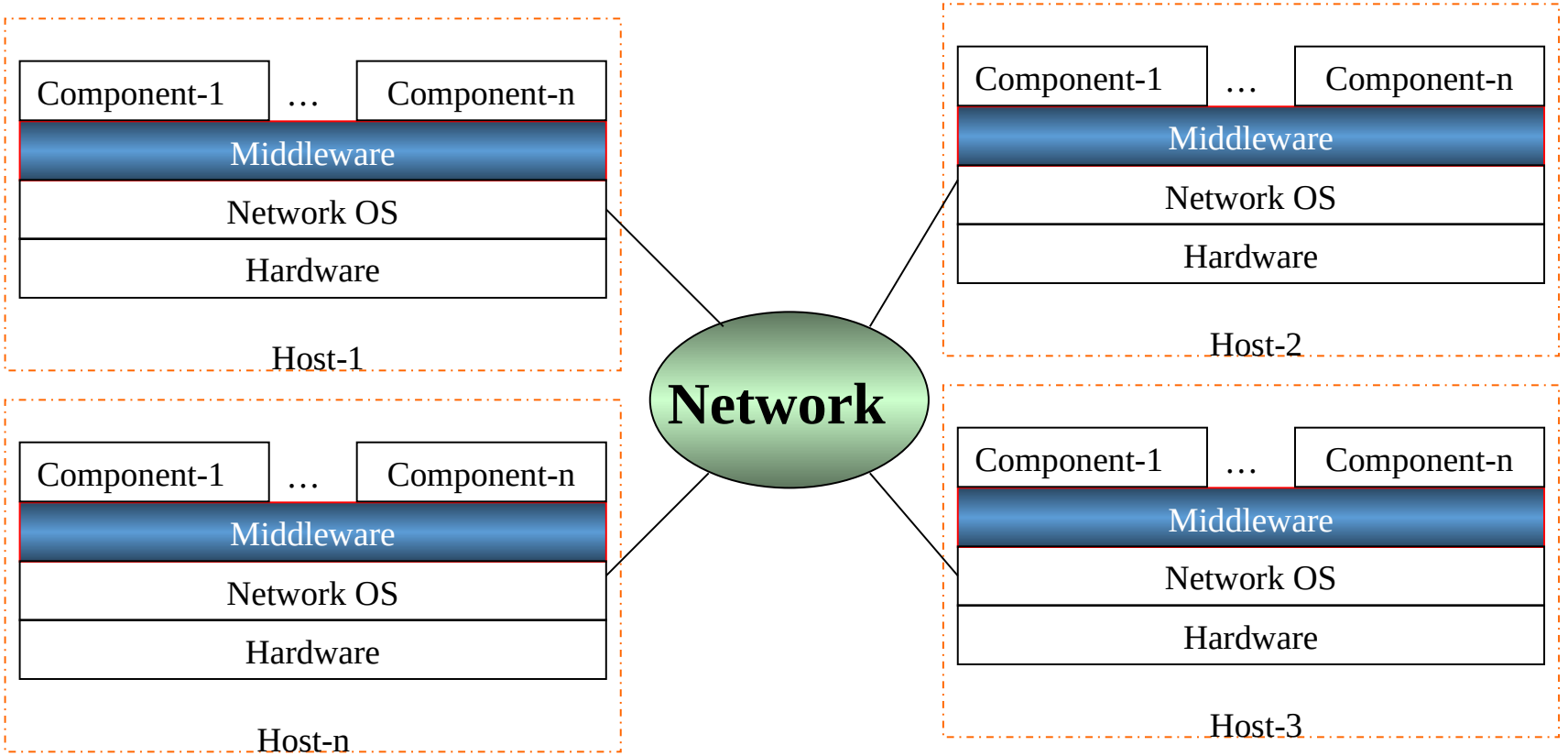
A Distributed Computer System is a system with:

1.Group of autonomous system

2.Geographically dispersed

3.Connected via a network

4.Having common/shared applications



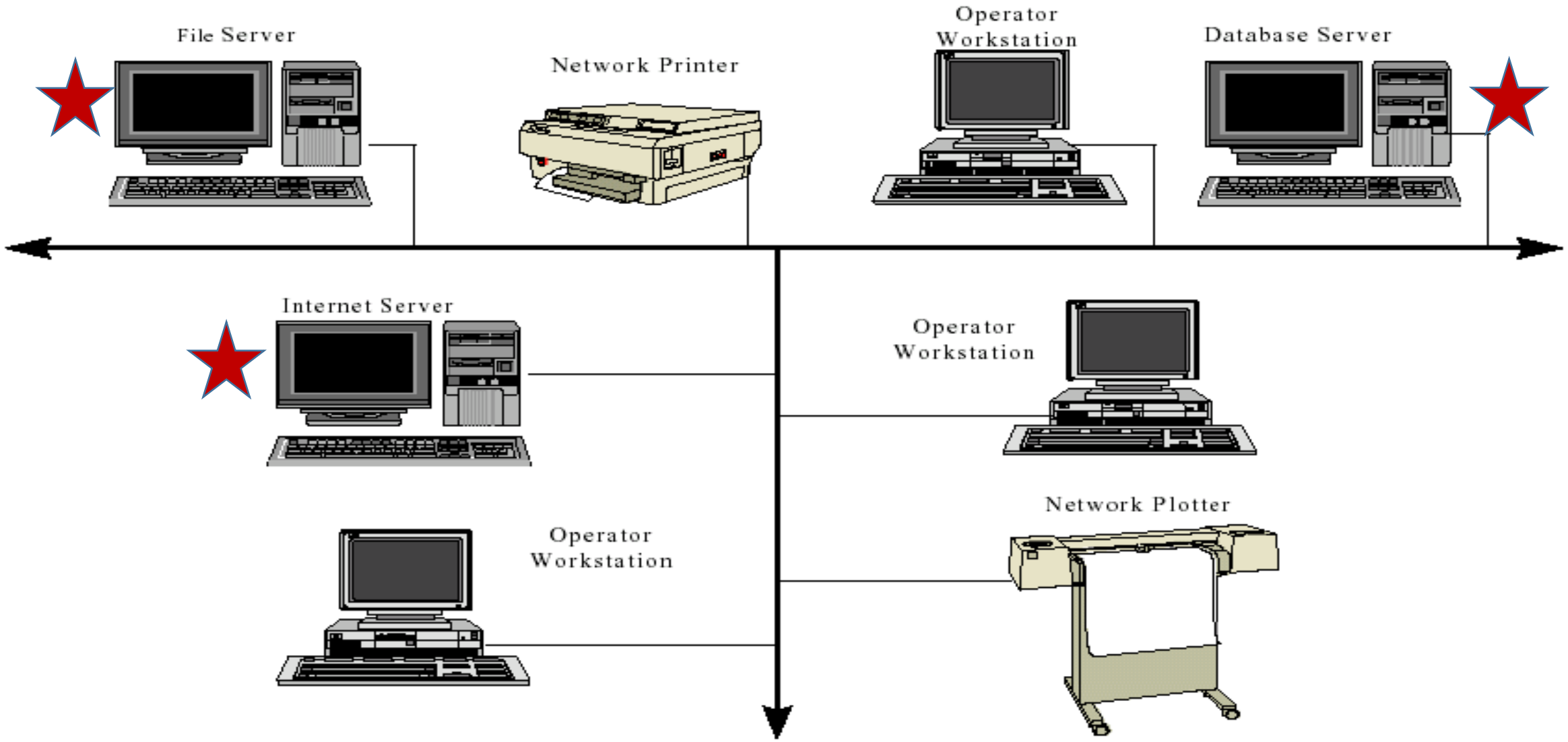
2- Geography Dispersed

3- connected Via Networks

4- having common applications

Components of Distributed Software:
Distributed operating systems
Middleware
Distributed applications

Distributed Computer System Architecture over a LAN



Distributed Servers: File, Applications, Database, Internet, Print,……..

Distributed Clients: Workstations and Personal computers